

Artificial neural networks in semiconductor device compact modeling: A review

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ABSTRACT

This review examines the use of artificial neural networks (ANNs) in the development of compact models for emerging devices. As new device technologies emerge, there is a critical need for compact models that are not only accurate and computationally efficient, but also sufficiently simple and physically consistent with support meaningful circuit-level simulation and design. We discuss how ANNs can be employed to address this challenge, emphasizing their potential to complement or extend traditional compact modeling approaches. The paper also outlines prospective research directions, focusing on the integration of machine learning techniques—particularly neural network architectures and training strategies—within the compact modeling framework. Finally, key methodological aspects are addressed, including the selection of training datasets, definition of model parameters, and relevant performance metrics, to guide the development of robust and interpretable models.

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I. INTRODUCTION

In the context of the fourth industrial revolution, the demand for electronics capable of producing, processing, storing, and transmitting data in a hyperconnected world is rapidly increasing. However, the development of semiconductor devices and integrated circuits (ICs) remains a complex and resource intensive endeavor, requiring substantial costs and long development cycles. The process typically begins with the conception and integrated circuit (IC) design phase, conducted by fabless companies, followed by the fabrication phase in foundries, and finally packaging and test phase in outsourced semiconductor assembly and test (OSAT) companies. Foundries provide fabless companies with process design kits (PDKs), which contain the essential information on the manufacturing process

technology. To address the challenges in this workflow, recent studies have reported promising applications of artificial intelligence (AI) in semiconductor design and manufacturing.¹

Artificial intelligence (AI) has demonstrated significant potential across the entire semiconductor integrated circuit (IC) design flow, from front-end to back-end stages. In front-end design, AI enhances functionality and performance by automating complex tasks such as architecture exploration, circuit optimization, and layout generation.² At the Register-Transfer Level (RTL), machine learning (ML) techniques have been proposed to directly evaluate Verilog-based RTL designs without requiring synthesis, enabling faster feedback and more efficient design space exploration.³ In back-end design, AI is increasingly used to tackle the intricate and time-intensive challenges of physical design, particularly in placement and routing.⁴ This is especially valuable in analog IC design, where conventional tools struggle with analog layout constraints.⁵ A prominent example is the application of reinforcement learning to chip floorplanning for **Tensor Processing Units (TPUs)**, which has replaced thousands of hours of manual effort while achieving competitive design quality.⁶ Additionally, transistor sizing in analog amplifiers has been optimized using a hybrid approach that integrates **graph convolutional neural networks (GCNs)** with reinforcement learning, treating the circuit as a graph and learning sizing decisions through iterative experience.⁷ As design closure and sign-off processes become increasingly complex, the role of AI in these final stages is expected to grow, offering greater scalability, predictability, and automation.⁸

In addition to their role in design, AI technologies are increasingly being integrated into semiconductor manufacturing to improve efficiency and reduce costs. Prior to fabrication, AI can assist in selecting optimal materials through intelligent material screening, enhancing the overall efficiency of the process.⁹ Recent studies have explored the use of AI for real-time monitoring and control of process parameters. For instance, AI-driven parameter learning has been applied to model chemical-mechanical polishing (CMP), enabling accurate predictions of thin-film profile heights and geometric characteristics.^{10,11} In lithography, deep learning algorithms have been used to simulate and predict critical dimensions with high accuracy and significantly reduced computational time.^{12–14} Additionally, inverse design techniques using AI have shown promise in predicting and optimizing device structures based on target characteristics, advancing toward practical fabrication.^{15,16} AI has also been applied to improve production yield through predictive algorithms,¹⁷ and to analyze complex patterns in large-scale datasets generated during semiconductor manufacturing.^{18,19} These developments underline the transformative potential of AI in streamlining and enhancing semiconductor production processes.

Compact models, which are sets of mathematical equations describing the behavior of electronic devices, serve as a critical bridge between device technology and circuit design.²⁰ However, developing and standardizing such models is a complex task due to the wide variety of device structures, material systems, and operational modes (DC, AC, and transient). In response to these challenges, recent research has explored the use of artificial intelligence—particularly the function approximation capabilities of neural networks²¹—in compact modeling, leading to the emergence of neural compact models.

Since the first demonstration of a neural network-based DC model,²² neural compact models have evolved significantly. A major milestone was the commercialization of DynaFET by Keysight

(formerly Agilent), which incorporated exact adjoint sensitivity analysis.²³ This advancement laid the groundwork for both DC and AC neural compact models in compound semiconductor devices. Notable examples include models for a 120 GHz GaN high electron mobility transistor (HEMT),²⁴ a 50 GHz GaAs HEMT,²⁵ GaN heterostructure field-effect transistor (HFET),²⁶ and 40 GHz GaN HEMT.²⁷ Leveraging diverse neural network architectures and advanced learning algorithms, neural compact models have shown continuous improvements in accuracy, scalability, and applicability across device types.

Despite this progress, there remains a lack of comprehensive reviews addressing the core components of neural compact modeling—such as network architecture design, training algorithms, computational cost, and evaluation criteria and methodologies. This review aims to fill that gap by presenting a detailed overview of the current state-of-the-art in neural compact modeling, while outlining key challenges and future research directions in this rapidly evolving field.

The remainder of this review is structured as follows. Section II provides foundational background, including an overview of conventional compact modeling and artificial intelligence. Section III focuses on neural compact models, detailing both shallow and deep neural network (DNN)-based approaches. Section IV discusses current challenges, outlines future research directions. Finally, Sec. V concludes the review with a summary of key insights and perspectives.

II. BRIEF OVERVIEW

A. Compact model

A compact model is a set of mathematical descriptions (equations) of semiconductor devices written in computer language and used in analog circuit simulator.²⁰ It provides an efficient way to predict how the circuit will behave before manufacturing it and to evaluate the circuit design with the measurement results for further design optimization, enabling significant reduction of circuit development cost and time. In order for the results of circuit simulation useful for a circuit designer, the compact model must be:

- Simple: to be able to design a circuit consisting of as many transistors as possible, at a reasonable computing power and time. Conventionally, this criterion has been addressed by mathematical approximation and by physically valid assumptions.
- Accurate: to ensure that the circuit is made as designed. The model should exhibit good accuracy over a wide range of device operation regime, i.e., DC and AC. The error in device-level modeling accumulates to the error in circuit simulation.
- Physically rigorous: to provide insights from simulation results for further advancement or optimization of the design. In order to do so, the model should be calibrated to experimental data and/or be based of physical equation.
- Timely: to keep up with the advancement of new technology, for example, recent transition from FinFET to Gate-All-Around-FET (GAAFET). The compact model should be available for a certain device technology to be ready for IC development.

Satisfying all of the aforementioned criteria is a demanding task. Therefore, a compact model is developed, accredited, and promoted by the collaboration of university, industry, and allied organizations, such as the Compact Model Coalition of the Silicon Integration Initiative (Si2), Inc.

The use of standard FET compact models may face two challenges: (1) emerging devices may display electrical characteristics that are not well captured by the standard FET models, and developing the physics-based model equations for the new physical phenomena requires high expertise and a long TAT (Turnaround Time) and (2) for equation-based models, it is still challenging to fully automate the model parameter extraction process while achieving a very high fitting accuracy.

B. Artificial intelligence

AI enables functional approximation of physical and mathematical functions as a convolution of weight and bias through the structure of artificial neural networks (ANNs). The use of AI for semiconductor design has been studied for decades. In the past, large-scale training was difficult due to issues such as data acquisition, computational speed, and storage space, so only simple predictions were performed on relatively simple structures such as those with one hidden layer. However, as technology has advanced, the aforementioned problems have been solved and the scale of AI has automatically increased. Therefore, there are the following essential considerations for AI learning to achieve a high level of accuracy by learning more complex neural network structures. AI model that achieves accuracy and reliability can only be considered “Good” if the items laid out in the criteria below are thoroughly and specifically taken into account.

- Data acquisition: to collect raw data from simulations, experiments, or real-world sources. High-quality data is the foundation of effective learning and ensures model expressiveness for the target task.
- Pre-processing: to normalize and clean the data using techniques like Min–Max scaling or Z-score standardization. This step ensures that features are on a comparable scale improving training efficiency. Additional denoising techniques can be applied together to handle noisy or inconsistent data.
- Neural network selection: to choose an appropriate architecture such as Multi-layer Perceptron (MLP), convolutional neural network (CNN), recurrent neural network (RNN), or Generative Adversarial Network (GAN) based on the task types and data characteristics. The chosen structure determines the model’s ability to capture spatial, temporal, or generative patterns in the data.
- Hyperparameter tuning: to optimize critical settings like learning rate, batch size, optimizer, and activation function. Proper tuning enhances convergence speed, stability, and generalization performance.
- Loss function: to use metrics such as mean squared error (MSE), mean absolute error (MAE) for regression or cross-entropy (for classification) to quantify prediction errors. The loss guides model training by showing how far predictions deviate from the ground truth.
- Optimization: to update model weights using algorithms like Gradient Descent, Adam, or RMSProp to minimize the loss. Optimization is the engine of learning, directly influencing model accuracy and efficiency.
- Model evaluation: to evaluate the trained model using accuracy, precision, recall, F1-score, root mean square error (RMSE) depending on the task. This step ensures the model generalizes well and performs reliably on unseen data.

Each component plays a crucial role in building reliable and generalizable AI-based compact models. Clean and well-balanced inputs enable efficient learning, while proper architectural choices and training strategies enhance the model’s ability to capture complex device behavior. Robust evaluation methods ensure that the model maintains accuracy and avoids overfitting, even on unseen data. When these essential elements are properly addressed, AI-augmented compact models can significantly advance the design and realization of next-generation devices. As shown in Fig. 1, this integrated framework supports the full pipeline from early-stage device modeling to practical deployment. In the face of increasing design complexity and the lack of closed-form solutions for novel architectures, leveraging AI offers a powerful and scalable solution for compact modeling in emerging technologies.

III. NEURAL COMPACT MODEL

Neural networks offer a universal function approximation framework ideally suited for compact modeling of semiconductor devices, enabling them to learn I – V , C – V , and other multiphysics response surfaces directly from data. As network architectures grow deeper and training algorithms become more sophisticated, these models are increasingly capable of capturing the intricate, nonlinear device behaviors that arise in advanced transistor and emerging device designs.

A. Neural compact model

Figure 2 presents a chronological timeline of the seminal developments in neural network-based compact modeling for semiconductor devices, tracing the field’s progress from its inception in the early 1990s to the latest deep-learning and graph-based frameworks.

The development of neural compact models has evolved significantly since the early 1990s, beginning with the introduction of the first DC neural compact model for MOS transistors.²² This conceptual foundation was soon extended to GaAs MESFETs with both DC and AC modeling,²⁸ initiating the diversification of neural modeling into high-frequency devices. By the late 1990s, applications had been proposed for microwave and VLSI components such as coplanar waveguides, spiral inductors, and interconnects using neural network techniques.^{29–34} A shift toward empirical learning and automated data acquisition was further established through the use of measured DC and AC characteristics for MOSFET modeling.³⁵

More recently, efforts have been directed toward improving modeling accuracy and broadening applicability to emerging devices. A neural-based DynaFET model has been developed for GaAs and GaN HEMTs,³⁶ while a technology computer-aided design (TCAD)-augmented CNN learning framework was introduced to enhance compact model training.³⁷ Neural models have also been shown capable of simulating logic circuits such as XOR and SRAM (Static Random Access Memory) with performance comparable to conventional BSIM-CMG models.³⁸ Furthermore, the use of graph-based architectures has enabled the creation of graph neural network (GNN)-based neural compact models with SPICE-compatible implementations, offering improved flexibility and structural awareness.³⁹ Collectively, these advancements reflect a transition from conceptual validation to scalable, high-accuracy modeling frameworks suitable for modern semiconductor technologies.

Conventional compact models for semiconductor devices have long relied on physics-based equations to predict behaviors such as

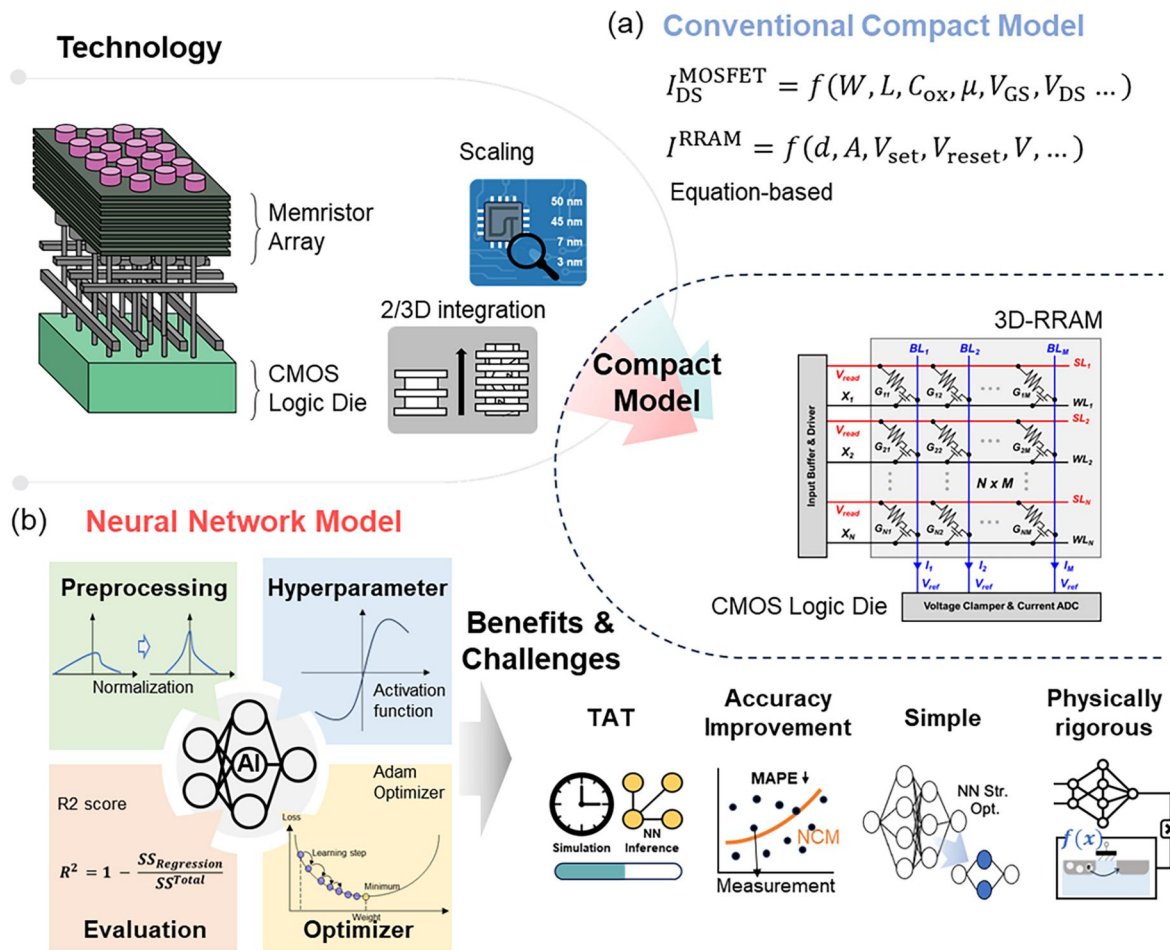


FIG. 1. One of various emerging technologies, i.e., AI accelerator based on memristor array and CMOS logic chips, which requires device compact models for both transistor and memristor for design and technology co-optimization. (a) Conventional approach based on a set of mathematical equations coded in computer language. (b) A new approach exploiting the ability of neural networks in function approximation and artificial intelligence.

I - V characteristics, C - V relationships, and impedance properties. These models, including industry standards like BSIM and BSIM-CMG (Berkeley Short Channel IGFET Model-Common Multi Gate), are rooted in semiconductor physics and often require the definition of many process-dependent parameters, along with extensive calibration. While they provide high accuracy and physical interpretability, their adaptability to nonstandard structures is limited such as flexible devices, 2D materials, or emerging memory technologies. Reconfiguring models for such novel devices often demands time-consuming derivation and empirical fitting. As devices become more complex and heterogeneous, these traditional methods struggle with generalizability and scalability.

Neural compact models provide a data-driven abstraction of semiconductor devices by training ANNs directly on simulation and experimental acquisition data. As shown in Fig. 3, key device inputs, such as material parameters, geometric dimensions and bias conditions, are fed into the network and mapped to electrical outputs such as DC I - V curves, C - V profiles, and S -parameters. Training combines

calculated and measured datasets, and a systematic hyperparameter fitting step (tuning layer depth, neuron counts, learning rates, and regularization strengths via grid search or Bayesian optimization) ensures both high fidelity and robust generalization. Once tuned, the neural compact model abstracts underlying physics without explicit analytic equations, delivering rapid, accurate predictions even for novel device architectures. Moreover, the approach scales from shallow MLPs to deep CNNs and graph-based networks capable of capturing intricate spatial and topological effects, heralding a flexible, data-centric paradigm for next-generation semiconductor design.

B. Shallow neural network based compact model

As mentioned before, ML-based compact models have emerged as a promising alternative by leveraging NNs trained on simulation data and experimental measurements. These data-driven models significantly improve modeling accuracy, scalability, and adaptability compared to traditional physics-based compact models, especially

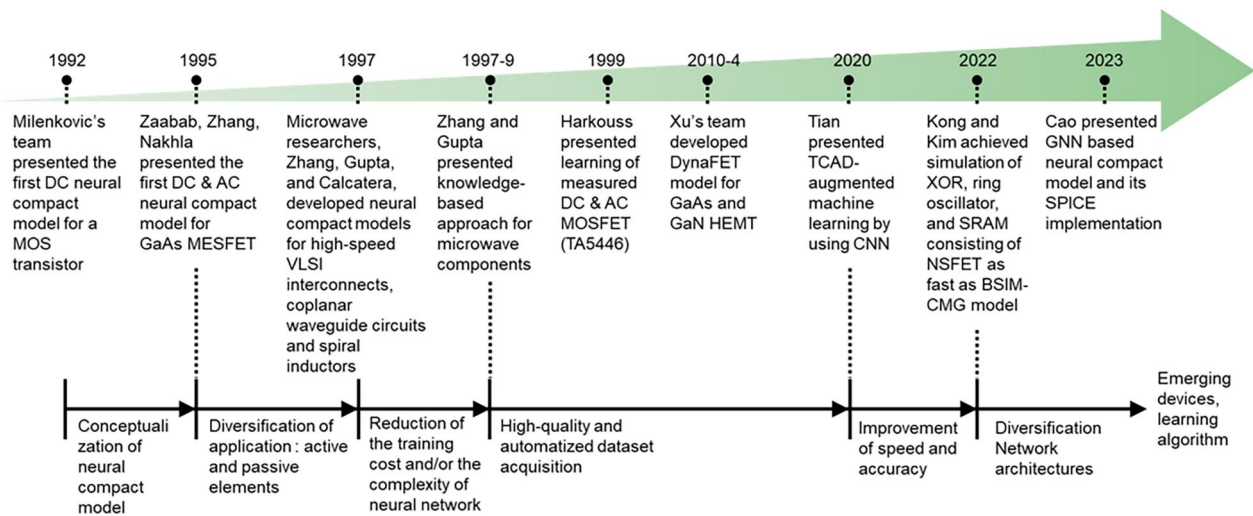


FIG. 2. Key milestones and perspectives in neural compact model research. Development of neural compact models for emerging devices could be accelerated by advanced neural network architectures and learning algorithms.

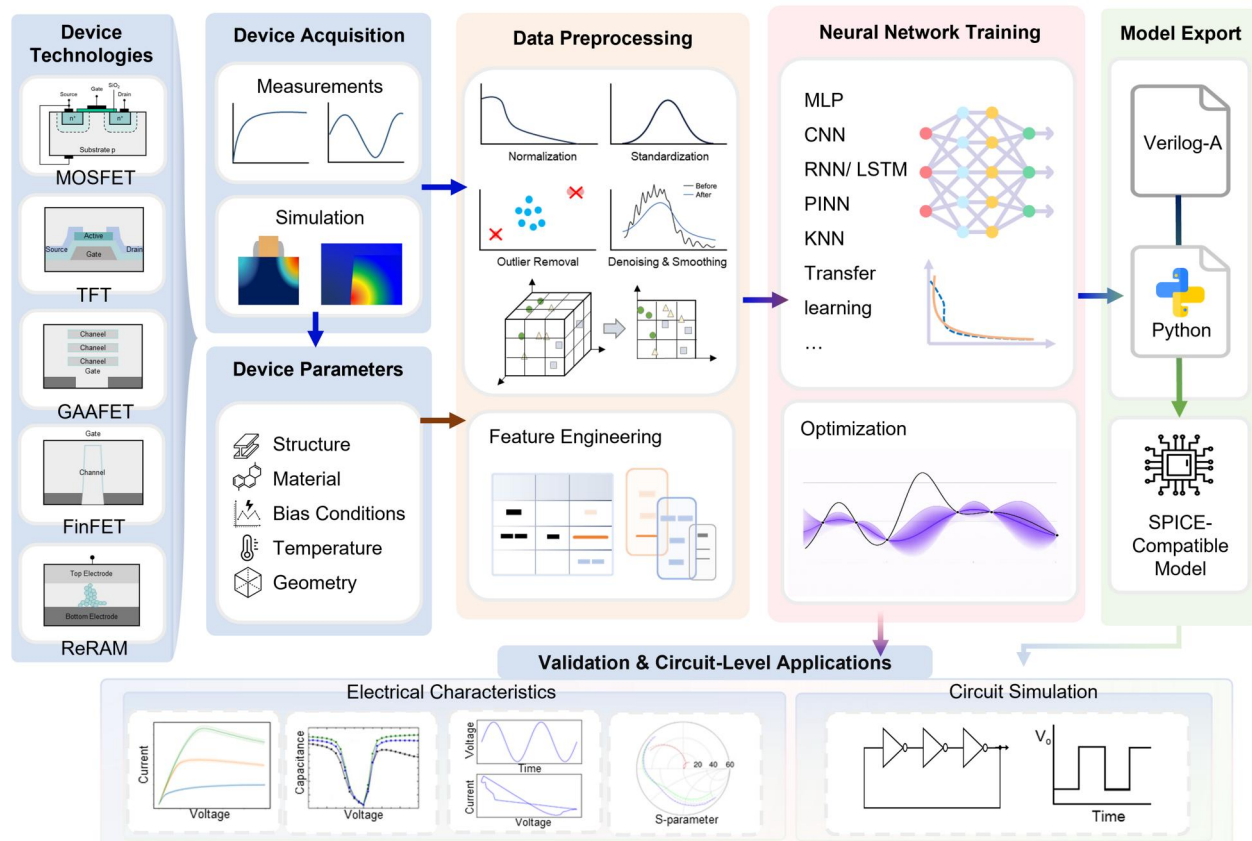


FIG. 3. A typical workflow in neural compact model development. Major steps are data acquisition by measurement or calibrated technology computer-aided design (TCAD), labeling (based on device structures, operation conditions, etc.), preprocessing of data, neural network model training, export, and circuit simulation, and validation. This diagram illustrates how neural networks are employed to develop SPICE (Simulation Program with Integrated Circuit Emphasis)-compatible compact models.

when applied to next-generation semiconductor technologies such as FinFETs, GAAFETs (Gate-All-Around Field Effect Transistor), and emerging memory devices. Among various NN architectures, shallow neural networks (SNNs) have been widely adopted due to their simple structure and relatively low computational cost.

This section reviews various SNN-based compact modeling techniques, with a particular focus on MLPs and their simplified variants. These models typically consist of a small number of hidden layers and moderate parameter sizes, making them well-suited for applications involving limited datasets or restricted computing resources. SNN-

based models are often employed to learn relatively simple or static device characteristics, such as DC I - V curves or low-frequency capacitance profiles, where the nonlinearities are not too complex and can be captured by shallow architectures. Their ease of implementation and training efficiency make them a practical choice for rapid modeling and prototyping in standard CMOS technologies.

However, as the complexity of semiconductor devices increases—due to factors such as multi-dimensional process parameters, temperature variation, and dynamic behavior—MLP-based models often exhibit limitations in capturing intricate nonlinear

TABLE I. Summary of shallow neural network-based compact modeling studies using MLP and its variants.

Additional algorithm	Device	Data acquisition (data size)	Circuit simulation	Device operation	References
...	GaN HEMT	Measurement (-)	Amplifier	DC/AC	24
	SOI MOSFET	SPICE (41 200)	...	DC	40
	14 nm FinFET, NC-FinFET	BSIM-CMG (-)	...	DC	41
	GAAFET, NC-GAAFET	TCAD (-)	CMOS inverter (17-stage)	DC	42
	GAAFET	TCAD (405)	XOR, RO ^a (17-stage), 6T-SRAM	DC/AC	38
Neuro Evolution Behavior-based model	GAAFET	TCAD (1104 datasets)	6T-SRAM	DC	43
	OFET	Measurement (100)	Inverter, AH ^b circuit	DC	44
	nMOSFET	Measurement (-)	...	DC	45
	SiC MOSFET	Measurement (-)	DPT ^c circuit	DC/AC	46
	IZO TFT ^g	Measurement (-)	Inverter	DC	47
PSO ^d , L-BFGS ^e	MOSFET	Silvaco Atlas (35)	...	DC	48
LM ^f	GAAFET	TCAD (over 3000 datasets)	RO (19-stage), Op-AMP (two-stage)	DC	49
Retargeting method	FinFET, GAAFET	TCAD, Measurement (-,-)	RO (17-stage)	DC	50
Feature generation	MOSFET	BSIM and Specter simulator (-)	...	DC	51
Physics-informed	DRAM, Logic, Emerging FETs	Measurement, TCAD (-,-)	...	DC	52
	12 nm GAAFET	TCAD (17 357)	...	DC	53
KNN	RRAM	Measurement, SPICE (-,-)	1T1R cell	Transient	54
	Si MOSFET	Si MOSFET: BSIM6	RO (five-stage), inverter	DC	55
	MoS ₂ FET	MoS ₂ FET: Measurement (-)	RO (17-stage), logic gate		
Bayesian Regularization	GFET	SPICE (3922)	Frequency multiplier	DC	56
	NAND, NOR	Measurement, TCAD (-,-)	NAND, NOR	DC	57
Log-transformation	Generic three-terminal transistor	Measurement, TCAD (-,-)	FinFET, MOSFET	DC	58

^aRO: Ring Oscillator.
^bAH: Axon-Hillock.
^cDPT: Double Pulse Test.
^dPSO: Particle Swarm Optimization.
^eL-BFGS: Limited memory Broyden Fletcher Goldfarb Shanno.
^fLM: Levenberg-Marquardt.
^gIZO: Indium Zinc Oxide.

relationships. To overcome these challenges, enhanced SNN models have been developed by integrating additional algorithmic strategies such as advanced optimization, evolutionary tuning, and regularization techniques. Furthermore, domain-specific knowledge has been embedded into the learning process through physics-informed loss functions or transfer learning schemes. These hybrid and assisted approaches have proven effective in improving model generalization, robustness, and physical consistency while preserving the interpretability and efficiency of shallow architectures.

To support the discussion on shallow neural network-based compact models, Tables I and II present a broad set of representative studies that demonstrate their applications and training strategies. Table I summarizes MLP-based compact modeling efforts, including both standard and enhanced variants that incorporate auxiliary algorithms. The table highlights the diversity of target devices as well as various data acquisition sources, and validation circuits used to confirm model reliability.

Table II provides detailed insights into the neural architectures and training configurations employed in these studies. It includes the number of hidden layers and neurons, activation and loss functions, optimizers, learning rates, and training epochs. In addition, it reports performance metrics and the computing environments used for model training. Together, these tables illustrate not only the versatility of SNN-based compact modeling but also the range of factors that influence model accuracy and efficiency. They serve as a comparative reference for understanding how architectural choices, algorithmic enhancements, and data strategies affect neural compact model development across a wide array of semiconductor technologies. Table III presents customized loss functions designed for each study, highlighting how model optimization was tailored to meet the specific requirements of different neural compact modeling tasks. In compact modeling, the choice of loss function plays a crucial role in guiding the learning process toward physically meaningful and simulation-compatible outcomes. Rather than using standard regression losses such as MSE alone, many studies incorporate additional terms into their objective functions to reflect domain-specific knowledge.

Figure 4 contrasts three NN compact model strategies—pure MLP, physics-informed, and knowledge-based—along four key dimensions. In the “MLP only” strategy, device parameters, such as bias voltages, geometry, material constants, pass directly through a fully connected (FC) network to reproduce DC I - V , C - V , S -parameter and transient response curves, with the resulting Verilog-A or SPICE model validated on circuits such as ring oscillators (ROs), inverters and amplifiers. The physics-informed approach embeds analytic device equations within the neural network to enforce physical consistency and achieve high accuracy. The knowledge-based method applies a teacher–student framework to distill and transfer approximate prior knowledge (APK) into the network. Both the MLP-only and physics-informed models have demonstrated circuit-level validity, while validation of the knowledge-based model remains under way. Together, these approaches span a continuum from fully data-driven simplicity through physics-grounded accuracy to knowledge-driven flexibility, offering a comprehensive toolkit for relatively simple semiconductor device design.

1. MLP-based neural compact model

In neural compact modeling, the relationship between input and output is approximated not through explicit physical equations, but

via weighted summation and nonlinear activation functions within neurons. This enables NNs to act as powerful function approximators, especially when trained on simulation or experimental datasets. One example applies a neural network solely to the current source (I_{ds}) modeling, while retaining traditional models for resistance, inductance, and nonlinear capacitance. This hybrid approach mitigates overfitting risks associated with directly learning from S -parameter data. In this study, the network consisted of three hidden layers trained with the Adam optimizer and sigmoid activation, and its performance was validated through RF circuit simulations for millimeter-wave GaN HEMT modeling.²⁴ For SOI MOSFETs, where floating-body effects significantly affect behavior, a neural network approach was compared against Generalized Radial Basis Function (GRBF) models. The use of Region Partitioning—separately training models for triode and saturation regions—enabled stable prediction performance across channel lengths and bias regimes, leading to enhanced accuracy.⁴⁰

In addition to measurement-driven models, simulation-only approaches have also been explored, where TCAD-generated datasets are exclusively used for training NN-based compact models. These methods provide flexibility in modeling emerging devices without the need for extensive physical measurements and are well suited for scaling across technology nodes. For example, a fully connected MLP with two hidden layers (each with 500 neurons), trained using PReLU activation and stochastic gradient descent (SGD), was applied to model 14 nm FinFET and negative capacitance (NC)-FinFET characteristics. The model achieved an R^2 score exceeding 0.99, indicating excellent agreement with TCAD data.⁴¹ Beyond transistor modeling, neural networks have also been utilized to model sensor behavior under nonlinear effects, temperature variation, and noise. One study introduced a backpropagation neural network with a single hidden layer for error correction, while another proposed a functional link neural network trained using least mean squares for improved robustness.⁴²

Hybrid approaches that combine TCAD simulations and experimental data have proven especially effective in enhancing both physical consistency and generalization. For instance, a neural compact model for sub-3 nm GAAFETs was developed using TCAD data, employing two hidden layers (20 and 15 neurons) and a physics-informed loss function based on I - V and C - V characteristics. This model achieved error rates of 2.5%, 2.0%, and 1.0% across OFF, medium, and high V_{gs} operating regimes, respectively, and required ~ 1 h of training on a standard CPU (Central Processing Unit).³⁸ Another sub-3 nm GAAFET model, trained on 1104 samples (595 training, 105 validation, and 404 test), used four hidden layers (128-64-32-32), ReLU activation, and MSE loss. It achieved a correlation coefficient greater than 0.99, demonstrating strong predictive performance across a wide range of process conditions.⁴³

Furthermore, a data-driven compact model for organic field effect transistors (OFETs) was proposed using a two-layer MLP (256-128) trained on I - V measurements from 100 fabricated devices.⁴⁴ The network, optimized using SGD and polynomial feature transformation of V_{gs} and V_{ds} , predicted log-transformed drain current values with R^2 scores above 0.999 in the best cases and over 0.98 consistently. The model required only two minutes to train using Scikit-Learn and was successfully integrated into Cadence Specter, enabling direct use in circuit-level simulations such as inverters and

TABLE II. Shallow neural network architectures and training strategies in compact modeling. Hyperparameters are listed in the following order: activation function, loss function, optimizer, learning rate, and epoch. T : Temperature, t : thickness.

Input	Output	Hidden layers (neurons)	Hyperparameters	Accuracy	Hardware (training time)	References
V_{gsi}, V_{dsi}	I_{DS}, I_{GS}	3 (–)	Sigmoid; –; Adam; –; –	Verified by measurement	N/A	24
V_{gs}, V_{ds}, L, W	I_d, g_m, g_d	MLP: 1(5–20), 2(5–10); GRBF: 1(25–100)	tanh, Gaussian; MSE; –; –; 50–200	MSE < 1%–2%	650 MHz Pentium III; 256 MB RAM (0.7–60 s)	40
$V_{gs}, V_{ds}, T, n_{fin}^b$	$I_d, C_{gd}, C_{gs}, C_{gd}$	2 (500–500)	PReLU; MSE; SGD; –; 400–500 ^a	R^2 _score up to 0.9999	– (few hours to days)	41
V_{gs}, V_{ds}	$I-V: I_d$ $C-V: Q_g$	$I-V: 3$ (8–8) $C-V: 2$ (15–10)	tanh; customization; Adam; 0.0001; –	Error < 1%–2%	Intel i9 (13.51 s)	42
$L_g, W_{sheet}, t_{sheet}$ V_{GS}, V_{DS}	I_d C_{gg}, C_{gd}, C_{gs}	$I-V: 2$ (20–15) $C-V: 2$ (10–5)	tanh; customization; Adam; 0.0001; up to 1 × 10 ⁶	$I-V$: Error < 1%–2% $C-V$: Error < 1%	– (1 h)	38
$I_{on}, I_{off}, V_{th}, SS$	PHIG ^c CDSC ^d U_0^e	4 (128–64–32–32)	ReLU; MSE; Adam; –; up to 40	Correlation > 0.99	Intel i3–8100 (3 s)	43
$V_{gs}, V_{ds}^2, V_{ds}, V_{ds}^2, V_{ds} \cdot V_{gs}$ W, L, V_g, V_d	$\ln I_d $ I_d	2 (256–128) 2–5 (<20 per layer)	ReLU; RMSE; SGD; –; 2000 Sigmoid, tanh; MSE, RMSE; GA, Adam; –; 1000	R^2 _score > 0.98 (best 0.999) log RMSE: 0.07–0.14	– (<2 min)	44
$I-V: V_{ds}, V_{gs}$ $C-V: V_{ds}$	$I_{ds}, C_{gd}, C_{gs}, C_{ds}$	$I-V: 1$ (10) $C-V: 1$ (1–2)	Sigmoid; RRMSE; –; –; – ^f	Accuracy: 1.5–3 × Angelow model	1.9 GHz Intel i7–8650U; 16 GB RAM	46
V_{ds}, V_{gs}, W	I_{ds}, g_m, g_d	2 (15–20)	Sigmoid, tanh; MSE; L-BFGS, PSO; –; –	MSE < 3%–5%	(–) N/A	47
V_{GS}, V_{DS}, W, L	I_{ds}, C_{gs}	3 (10–10–10)	Sigmoid; MSE; trainbr; –; –	MSE < 2%–5%	N/A	48
V_{gs}, V_{ds}	I_{ds}, Q_{gs}, Q_{gd} $g_m, dC/dV$	2 (<20 per layer)	tanh, ReLU; customization; SGD; –; –	MSE < 5%	– (5 h)	49
V_{gs}, V_{ds}, W, L, t	I_d, Q_g, Q_d, Q_s	Global: 2 (20–15) Local: 2 (10–5)	tanh; customization; SGD; –; 5000	MSE < 1%–2%	Global: – (<8 h) Local fitting: – (<1–2 h)	50
$V_{ds}, V_{gs}, V_{bs}, L$	I_{ds}, g_m	2 (52–26)	ReLU; percent loss; –; 0.0025; –	Error < 0.65% (I_{ds}) Error < 0.68% (g_m)	– (15–20 h)	51
$I_{ds}, I_g, V_{gs}, V_{ds}, T$ V_{gs}, V_{gd}	I_d, V_{th}, Q, g_m I_d, g_{ds}, g_m	N/A 3 (6–6–6)	ReLU; MAE; –; –; – tanh; relative error; Adam; 0.0001; about 100 000	Error < 5% Error < 1.3% (I_d) Error < 4.1% (g_{ds}) Error < 2.9% (g_m)	N/A N/A	52 53
V, W^{*g}	W^*, I_d	Linear: 1 (20) Log scale: 2 (20–20) Ideal: 2 (10–10)	tanh; (R)MSE, Cross-entropy; –; –; –	RMSE (discrete) = 0.000 125 RMSE	N/A	54

TABLE II. (Continued)

Input	Output	Hidden layers (neurons)	Hyperparameters	Accuracy	Hardware (training time)	References
V_{gs}, V_{ds}, W, L	I_{ds}, Q	N/A	...	(continuous) = 0.000 119	N/A	55
V_{gs}, V_{ds}, L, W	I_d	2 (15-12)	tanh; MSE; GA, LM; -;	Error < 1%–5% Error < 1%	N/A	56
VDD, V_{IN1}, V_{IN2}	IDD, V_{OUT}	2 (10-10)	Sigmoid; MSE; Bayesian; -; 1000	Error < 1%	N/A	57
V_{gs}, V_{ds}	I_d	2 (8-4)	tanSig, LogSig; MSE; LM; -; -	MSE < 0.2%	N/A	58

^aGRBF: Generalized Radial Basis Function.
^b n_{fin} : Number of fins.
^cPHIG: BSIM-CMG parameter for gate work function, V_{th} -related parameter.
^dCDSC: BSIM-CMG parameter for drain/source-to-channel coupling capacitance, SS (Subthreshold Swing)-related parameter.
^e U_0 : Low-field mobility.
^fGA: Genetic Algorithm.
^g W^* : State value.
^hPReLU: Parametric Rectified Linear Unit.
ⁱRRMSE: Relative Root Mean Square Error.

TABLE III. Customization loss function.

References	Loss function
42	$a \cdot \text{RMS} \left(\frac{y_{\text{true}} - y_{\text{pred}}}{y_{\text{true}}} \right) + b \cdot \text{RMS} \left(\frac{gd_{\text{true}} - gd_{\text{pred}}}{gd_{\text{true}} + \delta} \right) + c \cdot \text{RMS}(gm_{\text{true}} - gm_{\text{pred}}) + d \cdot \text{RMS} \left(\frac{\partial z_{\text{true}}}{\partial V_{DS}} - \frac{\partial z_{\text{pred}}}{\partial V_{DS}} \right),$
38	$a' \cdot \text{RMS}(y_{\text{true}} - y_{\text{pred}}) + b' \cdot \text{RMS} \left(\frac{\partial y_{\text{true}}}{\partial V_{GS}} - \frac{\partial y_{\text{pred}}}{\partial V_{GS}} \right) + c' \cdot \text{RMS} \left(\frac{\partial y_{\text{true}}}{\partial V_{DS}} - \frac{\partial y_{\text{pred}}}{\partial V_{DS}} \right) + d' \cdot \text{RMS} \left(\frac{\partial^2 y_{\text{true}}}{\partial V_{DS}^2} - \frac{\partial^2 y_{\text{pred}}}{\partial V_{DS}^2} \right)$
49	$\frac{1}{n} \sum_{i=1}^n \left\{ \alpha(y_{\text{true,off}} - y_{\text{pred,off}}^2 + \beta(y_{\text{true,medium}} - y_{\text{pred,medium}}^2 + \gamma(y_{\text{true,high}} - y_{\text{pred,high}}^2) \right\}$
50	$\frac{1}{n} \sum_{i=1}^n (Y_i - Y'_i)^2 + \sum_{j=1}^m \frac{a_m}{n} \sum_{i=1}^n \times (\text{grad}_i^m[j] - \text{grad}_i^{m'}[j])^2$ $\frac{1}{N_s} \sum_{i=1}^{N_s} \left[(1 - d1) \cdot \text{err}(M^{(i)}) + (d1 - d2) \cdot \text{err} \left(\frac{dM^{(i)}}{dV_{gs}} \right) + d2 \cdot \text{err} \left(\frac{dM^{(i)}}{dV_{ds}} \right) \right]$

artificial neuron circuits. This case demonstrates the feasibility of ML-based compact modeling in real-world EDA (Electronic Design Automation) workflows, offering high modeling fidelity with significantly reduced parameter tuning effort.

Recent studies further demonstrate that MLP-based neural compact models can serve as efficient surrogates for global compact model parameter extraction without iterative optimization. In particular, single-network MLP frameworks have been shown to directly extract both I–V and C–V BSIM-CMG parameters over a wide range of gate lengths for FinFET technologies, achieving fitting accuracy comparable to manual extraction while significantly reducing extraction time.⁶⁰ More recent extensions employ structured or multi-stage MLP architectures to decompose the extraction task into subthreshold, linear, saturation, and capacitance regions, enabling improved robustness and scalability across advanced GAAFET nodes.⁶¹ These approaches rely primarily on supervised learning with Monte Carlo generated datasets and highlight the maturity of MLP-based models as practical tools for compact model parameterization, particularly under DC operating conditions.

These studies collectively demonstrate the flexibility and scalability of shallow neural networks for compact modeling. By tailoring neural architectures and training strategies to device type, data availability, and target accuracy, researchers have developed lightweight yet robust models for a variety of semiconductor applications. As the

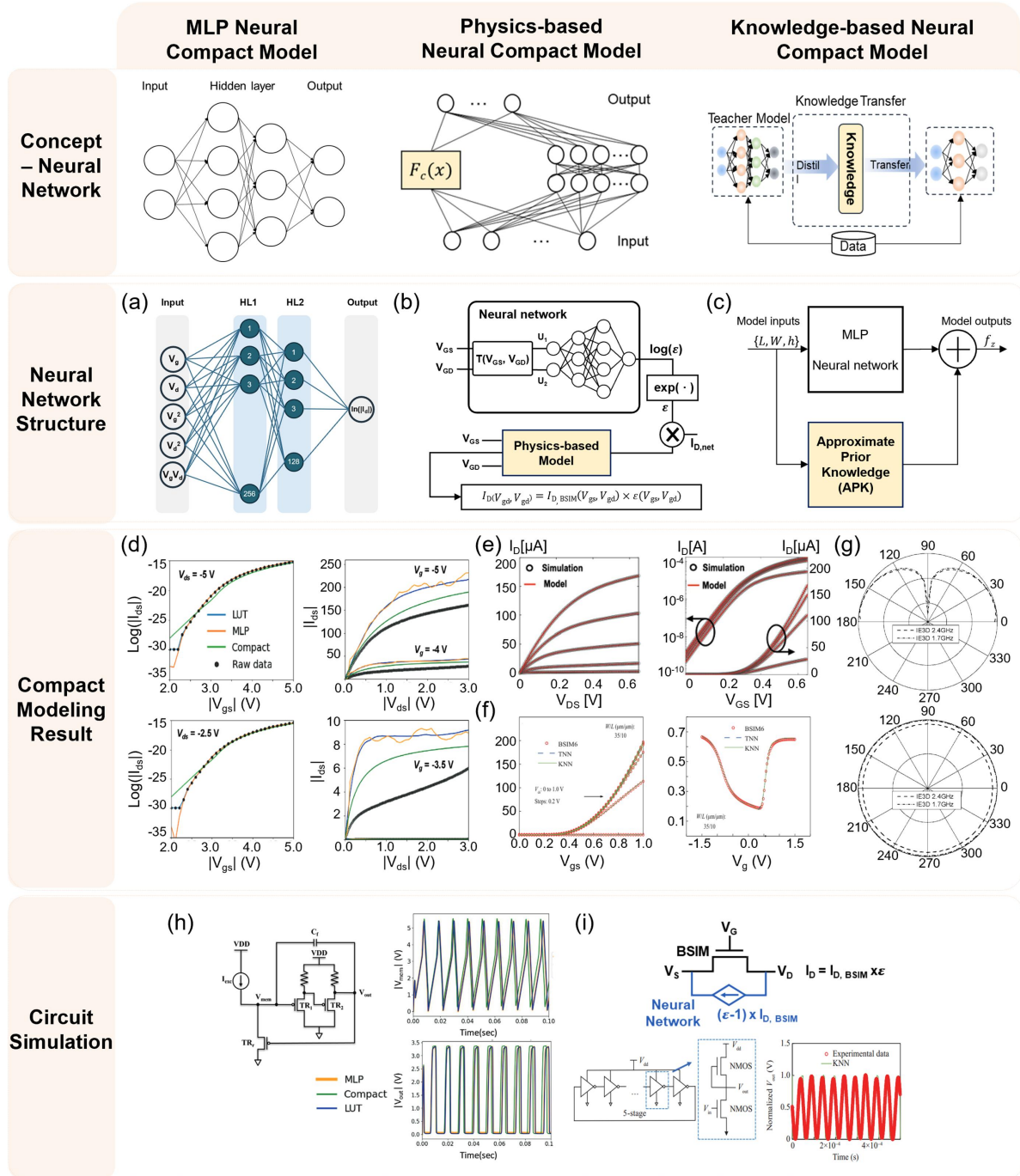


FIG. 4. Overview of shallow MLP-based neural compact models: (left) a standard MLP-only model, (center) a physics-based hybrid model, and (right) a knowledge-based model. The first row shows conceptual diagrams of each model type. (a) illustrates the architecture and application of a basic MLP model. Its compact modeling results for I–V characteristics⁴⁴ are shown in (d). (b) presents the physics-based model structure, where analytical functions are embedded alongside neural networks; corresponding I–V results⁵³ are shown in (e), and additional modeling outcomes⁵⁵ are visualized in (f). (c) displays a knowledge-based structure using Approximate Prior Knowledge (APK) integrated with an MLP, with validation results⁵⁹ provided in (g). Post-training circuit-level simulation results using each model are demonstrated in (h) and (i), validating the applicability of neural compact models in practical SPICE simulations. (a), (d), and (h) Adapted with permission from Calvet *et al.*, Adv. Electron. Mater. **10**, 2400515 (2024). Copyright 2024 Authors, licensed under a Creative Commons Attribution (CC BY 4.0) license. (e) Reproduced with permission from Kao *et al.*, IEEE Electron Device Lett. **43**, 974–977 (2022). Copyright 2022 IEEE. (g) Reproduced with permission from Xiao *et al.*, Int. J. RF Microwave Comput.-Aided Eng. **13**, 316–320 (2003). Copyright 2003 Wiley Periodicals, Inc. (f) and (i) Reproduced with permission from Qi *et al.*, Sci. China: Inf. Sci. **66**, 122405 (2023). Copyright 2023 Springer Nature.

field continues to evolve, the integration of domain knowledge, advanced optimization, and hybrid data sources will play a central role in bridging the gap between physical interpretability and data-driven adaptability.

2. Optimization-augmented MLP-based neural compact models

To improve training efficiency and predictive robustness even within simplified MLP architectures, a variety of assistive techniques have been explored. These include hybrid modeling approaches that incorporate domain-specific knowledge, mathematical constraints, or physics-based priors into the learning process. Optimization algorithms such as genetic algorithms (GA), Bayesian inference, and swarm intelligence have also been employed to tune model parameters or initialize weights more effectively. These strategies allow SNNs to retain computational efficiency while enhancing learning performance—particularly in scenarios where deep architectures may be unnecessary or impractical. In the following, we review studies that have applied such optimization methods to MLP-based compact models, highlighting their motivations and resulting improvements in modeling accuracy, generalization, and training efficiency.

One notable example applied Neuro-Evolution (NE), in which a GA was used to automatically optimize the topology of an MLP-based compact model. This method was introduced to overcome the limitations of manually designed networks, particularly in compact modeling scenarios requiring minimal model size and high accuracy. By jointly evolving the network structure and hyperparameters, the NE-based approach significantly reduced RMSE in MOSFET modeling while preserving a lightweight architecture suitable for circuit-level simulations. The model achieved a low log RMSE ranging from 0.07 to 0.14, demonstrating strong fitting performance across a wide range of operating regimes.⁴⁵

To mitigate modeling errors in the cutoff and low-voltage regions of MOSFETs, another study proposed a hybrid model that combined behavior-based equations with a lightweight ANN. The ANN handled nonlinear behavior in the active region, while empirical equations provided robust predictions in regions where purely data-driven models tended to overfit or extrapolate poorly. Implemented in the SIMetrix SPICE simulator and validated against double-pulse test (DPT) measurements, this hybrid approach achieved 1.5–3× higher accuracy and over 30% faster simulation time compared to conventional models. The neural network component employed a single hidden layer with fewer than ten neurons for I - V modeling and fewer than two neurons for C - V modeling.⁴⁶

In the context of metal-oxide thin film transistors (TFTs), a hybrid optimization strategy combining centroid opposition-based particle swarm optimization (COBPSO) with the L-BFGS algorithm was adopted. COBPSO optimized the initial weights to overcome L-BFGS's sensitivity to initialization and avoid local minima, while L-BFGS handled fast convergence in the final tuning stage. This method improved the accuracy of both large-signal (I - V) and small-signal (g_d , g_m) characteristics beyond what standalone MLPs could achieve. The resulting compact model was implemented in Verilog-A and validated through circuit-level simulations involving double-stage inverters and scan drivers.⁴⁷

For emerging TFT devices, an ANN-based compact model was developed to learn both drain current and gate capacitance directly

from TCAD simulation data. The model architecture consisted of three hidden layers with ten sigmoid neurons each, trained via Bayesian regularization. By taking bias voltages and device dimensions as inputs and outputting I_{DS} and C_{GS} , the model achieved high generalization capability even on unseen devices. Its smooth and differentiable output made it suitable for SPICE-compatible simulations, while its accuracy in modeling both nonlinear current and voltage-dependent capacitance demonstrated the advantage of data-driven approaches for devices with complex or unknown physics.⁴⁸

To further extend the modeling capability, a multigradient neural network (MNN)-based compact model was introduced for GAA Si cold source FETs. This model captured not only I - V and Q - V behavior but also higher-order derivatives such as G - V and C - V . Using over 3000 TCAD samples, the model was trained via a multigradient-aware backpropagation algorithm with SGD and momentum. It employed four hidden layers of 20 neurons with tanh activation and achieved mean squared errors below 5% for high-order derivatives. Compared to the weeks-long training typical of lookup-table (LUT)-based approaches, this MNN model converged within hours and was seamlessly integrated into SPICE simulations of circuits such as ring oscillators and operational amplifiers (Op-AMPs).⁴⁹

To handle variability across device instances, a global vs local fitting framework was introduced. A global MLP model (20–15 neurons) was trained across device corners, while lightweight local models (10–5 neurons) were generated for specific devices using a retargeting method. Follow-up enhancements incorporated custom loss functions including terminal derivatives, logarithmic current scaling for wide-range fitting, and voltage smoothing to satisfy symmetry conditions. Pretrained weights from similar devices reduced training time by 40–100×, and additional “PV (Process Variation) neurons” allowed modeling of process variation. These improvements enabled accurate Verilog-A-compatible models with less than 1% error in circuit-level metrics such as ring oscillator delay, outperforming conventional LUT-based approaches in flexibility and simulation efficiency.⁵⁰

Finally, a higher-order functional link ANN (FLNN) approach was adopted for g_m/I_d -based analog design. The model used two hidden layers (52–26 neurons with ReLU activation) and 22 higher-order transformed input features derived from four terminal voltages and dimensions. To manage the wide dynamic range of I_{ds} and g_m , training data were partitioned into four subregions, each trained with a percent-error-based loss function. Each ANN was trained on 2.5×10^6 samples within 15–20 min using PyTorch. The average prediction error remained below 1% for most cases, with worst-case deviation under 15%. In total, 60 ANN models were trained across various PTC (process, temperature, and corner) conditions, requiring just 740 kB of storage per output—more than 1000× smaller than traditional LUT-based models—making this approach highly practical for compact analog design workflows.⁵¹ Beyond direct parameter extraction, recent optimization augmented MLP-based neural compact models introduce auxiliary networks to enhance parameter flexibility while maintaining fast SPICE level evaluation.⁶² A representative example employs a two-stage architecture in which a parameter generation network programs the weights and biases of a primary MLP, allowing device technology parameters to be explicitly embedded into the model representation. This framework enables accurate modeling of large device ensembles while mitigating the typical trade-off between network size, accuracy, and simulation speed. By decoupling

parameter adaptation from current prediction, such architectures extend conventional MLP-based compact models toward more scalable and technology aware formulations, particularly suitable for design technology co-optimization (DTCO) workflows.

These optimization-augmented MLP-based compact models underscore the evolving role of shallow neural networks in delivering scalable, accurate, and resource-efficient modeling solutions. By combining algorithmic advancements with physical insight and targeted optimization strategies, these approaches provide a viable path toward next-generation compact models that meet both performance and design automation requirements.

3. Physics-informed neural compact model

Recent efforts in neural compact modeling have increasingly focused on incorporating physical knowledge into neural network training—an approach broadly referred to as physics-informed or physics-augmented learning. Unlike purely data-driven models, which often generalize poorly or violate known physical laws, physics-informed models embed domain-specific constraints into the learning process. This integration not only improves model interpretability and extrapolation behavior but also ensures compatibility with circuit simulation environments where physical consistency and differentiability are critical. Such frameworks are especially valuable in semiconductor device modeling, where accurate prediction of terminal characteristics and their derivatives is essential for analog, RF, and transient circuit analysis.

To enhance physical consistency and generalization in neural compact models, a physics-augmented modeling framework was proposed. **This approach incorporated physical loss functions based on device theory, such as temperature-dependent relationships for carrier mobility and Fermi potential, to constrain neural network behavior. By embedding these domain-specific constraints into the training objective, the model avoided unphysical inferences common in black-box models trained on sparse temperature data.** High-order derivative terms were also added to the loss function to ensure accurate prediction of small-signal parameters and SPICE compatibility. Weight decay regularization was applied to suppress overfitting to measurement noise, improving robustness for digital and analog circuit analysis. The resulting model achieved accurate and physically consistent predictions across unseen operating conditions, including temperature sweeps, and demonstrated scalability across advanced DRAM and logic technologies. Compared to traditional approaches, this method enabled interpretable, charge-conserving SPICE models with both high accuracy and technology independence.⁵²

To overcome the limitations of purely data-driven models, which often lack physical interpretability, a deep-learning-assisted, physics-driven compact modeling framework was proposed. This hybrid approach augments a simplified BSIM-CMG core with a trainable correction function $\varepsilon(V_{GS}, V_{DS})$ learned by a neural network. The correction term ensures symmetry, smooth differentiability, and improved modeling of non-ideal behaviors, while maintaining physical consistency such as $I_D = 0$ at $V_{DS} = 0$. A novel input transformation guarantees Gummel symmetry, and the training loss incorporates errors in I_D , g_m , and g_{ds} . The resulting model, trained on TCAD data of 12 nm GAAFETs, achieved 3σ prediction errors of 1.3% (I_D), 4.1% (g_{ds}), and 2.9% (g_m), and can be implemented in SPICE via an additive current source term. This method reduces

parameter extraction overhead, improves runtime efficiency, and offers strong compatibility with standard circuit simulation workflows.⁵³

A hybrid physics-augmented neural modeling approach was also proposed for resistive RAM (RRAM) compact modeling, particularly targeting transient behavior during set/reset operations.⁵⁴ The model architecture mimics the structure of physics-based models by separating the learning of the state variable W or its derivative dW/dt , and the output current I . Among several variants, a two-block MLP model was designed, where the first block predicts the device state and the second block predicts the current. To capture transient phenomena, the voltage sweep speed (rise/fall time) was introduced as an additional input, enabling the model to learn time-dependent switching behavior. This physical hybridization not only improves generalization but also enhances convergence stability during SPICE-level transient simulations. A classification-based model for discrete state prediction demonstrated superior accuracy with an RMSE of 0.000 125 on test data, while also achieving better numerical stability than continuous dW/dt -based models, which suffered from instability under adaptive time stepping. By combining a compact architecture comprising just two hidden layers with a structure grounded in physical modeling principles, the approach provides a scalable and SPICE-compatible solution for RRAM modeling in advanced memory design.

Recent studies suggest that physics-informed neural compact models provide an effective compromise between purely data-driven fitting and conventional physics-based formulations. The PHIMONN (Physics-Informed Modeling Neural Network) approach⁶³ embeds compact model core equations with lightweight neural networks to replace empirical fitting and binning parameters, enabling efficient one-shot parameterization while maintaining global consistency across device geometries. This strategy has been shown to significantly reduce parameter extraction time and to facilitate technology transfer from FinFET to nanosheet GAAFET without extensive reformulation. Another experience, physics-informed neural networks have been applied to memristor compact modeling by incorporating governing differential equations directly into the training process, allowing hysteresis and state dynamics to be captured in a physically consistent manner.⁶⁴ By enabling direct Verilog-A implementation, these approaches demonstrate circuit-level compatibility while improving model robustness and interpretability. Collectively, these results indicate that physics-informed neural compact models are emerging as a viable framework for scalable and simulation-ready modeling of advanced and emerging devices.

Together, these studies highlight the growing maturity and flexibility of physics-informed neural modeling strategies for compact device modeling. By embedding physical knowledge into neural network architectures, training objectives, or hybrid structural designs, these methods overcome many of the key limitations faced by black-box models, including poor extrapolation, lack of interpretability, and difficulty integrating into simulation workflows. As semiconductor devices continue to evolve and diversify, such hybrid approaches will play an increasingly essential role in ensuring accurate, generalizable, and simulation-ready compact models for next-generation electronic design.

4. Knowledge-based neural compact model

While traditional physics-based models offer strong interpretability, their complexity, and computational overhead often limit

scalability and adaptability—particularly for novel device architectures. In contrast, deep learning approaches provide rapid learning and flexibility but frequently suffer from poor generalization outside the training domain. To address these limitations, knowledge-based neural networks (KNNs) have emerged as a promising middle ground, embedding physical laws into the learning process to improve both predictive accuracy and physical consistency. By integrating prior knowledge as constraints or structural components of the neural architecture, KNNs are able to model complex device behavior while avoiding nonphysical extrapolations.

A representative application of this approach is the KNN compact modeling framework proposed for MOSFETs and 2D material devices such as MoS₂.⁵⁵ In this method, geometric parameters like channel width and length (W , L) are handled via analytical expressions based on physical theory, while the voltage-dependent characteristics are learned by the neural network. This separation of knowledge domains enhances model generalization, particularly for unseen device geometries. Compared to conventional black-box neural networks, which exhibited errors exceeding 20% in extrapolation scenarios, the KNN model reduced prediction error to below 8%. The model also successfully captured both DC and transient behavior in fabricated MoS₂ devices, enabling accurate process variation analysis and circuit-level simulation via SPICE-compatible Verilog-A implementations.

In a similar effort, a prior knowledge input (PKI) neural network was introduced for graphene FET (GFET) modeling, incorporating a simplified empirical current expression as an additional input to the ANN.⁵⁶ This prior knowledge—based on a hybrid of the Shichman–Hodges model and GFET-specific conduction characteristics—helped the network efficiently learn complex behaviors from limited data. The model, consisting of two hidden layers (15–12), was trained using adaptive genetic algorithms combined with Levenberg–Marquardt (LM) optimization and Bayesian regularization. It achieved over 96% accuracy within 1% prediction error on test data, even for devices outside the training range, and maintained robust performance with only one-third of the original training dataset. Furthermore, in circuit-level simulation of a GFET frequency multiplier, the model demonstrated a relative output error below 1%, underscoring its scalability and SPICE compatibility. Recent studies have further advanced knowledge-based neural compact models by explicitly embedding prior physical or low-fidelity model information into the learning process to improve both data efficiency and simulation performance. For advanced nanoscale devices such as nanosheet FETs,⁶⁵ knowledge-based neural networks have been shown to effectively combine TCAD data with compact models, significantly reducing the required number of TCAD simulations while maintaining prediction accuracy across global and local variation spaces. Other case, knowledge-based ANN architectures for FinFET compact modeling have demonstrated that incorporating analytical model outputs as prior knowledge enables more compact network structures, faster convergence, and substantially improved SPICE runtime compared with conventional ANN based compact models, while retaining TCAD level accuracy.⁶⁶

These studies illustrate how KNN-based compact modeling techniques can effectively bridge the gap between physics-driven and data-driven approaches. By embedding prior knowledge into neural architectures, researchers have developed models that are not only

accurate and data-efficient but also physically meaningful for emerging semiconductor technologies where data scarcity and interpretability remain key challenges.

5. Circuit level neural compact model

Circuit-level compact modeling extends this approach by modeling the entire behavior of functional blocks, such as logic gates or analog subcircuits, often incorporating nonlinear interactions between multiple device elements. This is particularly relevant for monolithically integrated circuits or mixed-mode systems where transistor-level modeling introduces cumulative errors and calibration complexity. Traditional empirical models rely heavily on the accurate modeling of each component transistor, followed by assembling them into circuits—a process that can amplify small modeling inaccuracies and limit scalability. To overcome this, NN-based approaches are gaining attention as they directly model the input–output behavior of the entire circuit block, abstracting the transistor-level complexity while retaining high prediction fidelity.

A NN-based compact modeling framework at circuit level was proposed for monolithic InAlN/GaN NAND and NOR logic gates, aiming to improve accuracy and simplify calibration.⁵⁸ Rather than constructing models from individual transistor components, the approach directly learns the large-signal behavior of the logic gates—including both current consumption (I_{DD}) and output voltage (V_{OUT})—using a feedforward neural network with two hidden layers trained via Bayesian regularization. This eliminates the need for manual calibration of each transistor, avoiding compounded errors from device-level variations. To further refine the model, nonlinear capacitances between input/output/supply nodes were also modeled using separate neural networks, trained on data from 2D FEM (Finite Element Method) simulations. This allowed accurate prediction of transient behavior and parasitic effects, which are critical for high-frequency digital applications. Compared to traditional transistor-based models, the NN approach reduced the average modeling error by approximately one order of magnitude, achieving average V_{OUT} and I_{DD} errors below 1%. Furthermore, the method demonstrated strong agreement with measured data and proved suitable for automated calibration and SPICE-level circuit simulation, making it particularly valuable for emerging GaN-based digital ICs.

A neural network-based compact modeling methodology was proposed to efficiently describe generic transistors using a standardized ANN architecture, focusing on both modeling accuracy and SPICE compatibility.⁵⁸ To ensure smooth extrapolation across all operating regions, including subthreshold, linear, and saturation, the authors introduced a log-transformation for input and output variables (V_{ds} , I_{ds}), which significantly improved the fitting accuracy near $V_{ds} = 0$ and enabled robust convergence in SPICE transient simulations. A sparsely sampled dataset, patterned to reflect physical regions of operation, further reduced training cost while preserving model generalization. Based on empirical analysis, a four-layer network topology (2-4-4-1) was identified as an optimal structure for generic three-terminal transistor modeling. To prevent overfitting and enhance smoothness of derivatives such as transconductance (g_m) and output conductance (g_d), the network was trained using Bayesian regularization. The proposed framework achieved low MSE (below 0.2%) while maintaining physical consistency, and was successfully applied to FinFETs with varying channel lengths and to SPICE-level

circuit simulations including CMOS inverters and 17-stage ring oscillators. This work presents a standardized ANN-based compact modeling methodology that effectively balances simplicity, accuracy, and physical consistency for modern transistor technologies.

C. Deep neural network based compact model

SNN-based neural compact model, as mentioned before, has proven effective in approximating device behavior for compact models under relatively simple conditions, particularly when the mapping between inputs and outputs is moderately nonlinear and well-bounded. These architectures are effective for fitting $I-V$ or $C-V$ characteristics under limited conditions, particularly when only a few physical parameters influence the output. However, when device behavior becomes more complex due to factors such as nonlinear interactions, high-dimensional parameter spaces, or temporal dynamics, shallow networks

often struggle to fully capture the underlying physical dependencies and variability in behavior.

To overcome these limitations, researchers have increasingly adopted deeper and more specialized neural network architectures. CNNs are useful for extracting spatial correlations and localized features from input data, which enhances prediction accuracy in cases where device characteristics are influenced by layout-dependent or spatially varying parameters. Recurrent neural networks (RNNs), including long short-term memory (LSTM) models, have proven effective in modeling time-dependent behaviors. These are especially relevant for devices like RRAM, where current-voltage responses vary with time and transient behavior must be learned accurately. By expanding the capacity to represent dynamic and nonlinear patterns, these architectures allow for more robust and generalizable modeling.

TABLE IV. Summary of deep neural network-based compact modeling studies using MLP and its variants.

Algorithm	Device	Data acquisition (data size)	Circuit simulation	Device operation	References
CNN	FinFET, MOSFET	TCAD (3000–20 000)	...	DC/AC	37
Transfer learning	OFET	TCAD, Monte Carlo (10 000)	Inverter	DC	67
Transfer learning	32 nm MOSFET, 45 nm MOSFET	SPICE (402)	...	DC	68
LSTM	RRAM	Measurement, process variation (–,–)	...	Transient	69
Autoencoder	FinFET	TCAD (250)	...	DC	70

TABLE V. Deep neural network architectures and training strategies in compact modeling. Hyperparameters are listed in the following order: activation function, loss function, optimizer, learning rate, and epoch. T : Temperature.

Input	Output	Hidden layers (neurons)	Hyperparameters	Accuracy	Hardware (training time)	References
$I - V$	$I - V$	1D Conv. + FC (–)	ReLU, tanh, Sigmoid; MSE; LeNet-4-based; –; 2000	$R^2_score = 0.999$	– (few hours)	37
$V_{gs}, V_{ds}, V_{bs}, W, L, T$	I_d	N/A	–; relative error; –; –; –	Error = 4.3%	– (646 s)	68
V_{gs}, V_{ds}, T	I_d	3 FC–2 conv.–3 FC	tanh, Sigmoid, ReLU; MSE; Adam, RMSProp; 0.001; 200	Error = 0.0226; $R^2_score > 0.998$	Intel i9-12900K, RTX 4090 (about 5–7 h)	67
V_{gs}, V_{ds}, T	I_{HRS}, I_{LRS}	Sinusoidal: 1 (100); RandomWalk: 1 (100); Anneal: 1 (200)	tanh; RMSE; Adam; –; 2000	Sinusoidal: RMSE = 0.0096; Process-aware: RMSE = 0.0028; RandomWalk: RMSE = 0.0148	– (<5 m)	69
$I - V, C - V$	$I - V, C - V$	5 (40-40-3-40-40)	ReLU; MSE; Adam; –; 500	I_{ON} ; $R^2_score = 0.99$; DIBL, g_m , I_{OFF} ; $R^2_score = 0.98^a$	– (<1 m)	70

^aDIBL : Drain Induced Barrier Lowering.

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In addition, techniques such as transfer learning and domain adaptation have been employed to address challenges related to limited data availability and to facilitate model reuse across different device types or process technologies. These approaches enable deep neural networks to retain and adapt previously learned knowledge, thereby reducing the need for extensive retraining and improving scalability. Building on these advances, Sec. III C introduces a range of recent studies on deep neural compact modeling, highlighting how

various architectures and learning strategies have been applied to enhance prediction accuracy, generalization, and circuit-level compatibility.

Similar to the presentation of SNN-based models, detailed information on DNN-based compact models is provided in Tables IV and V. These tables present detailed configurations of DNN architectures, including CNNs, RNNs, and Autoencoder (AE), applied to various device types and electrical characteristics. Each entry includes

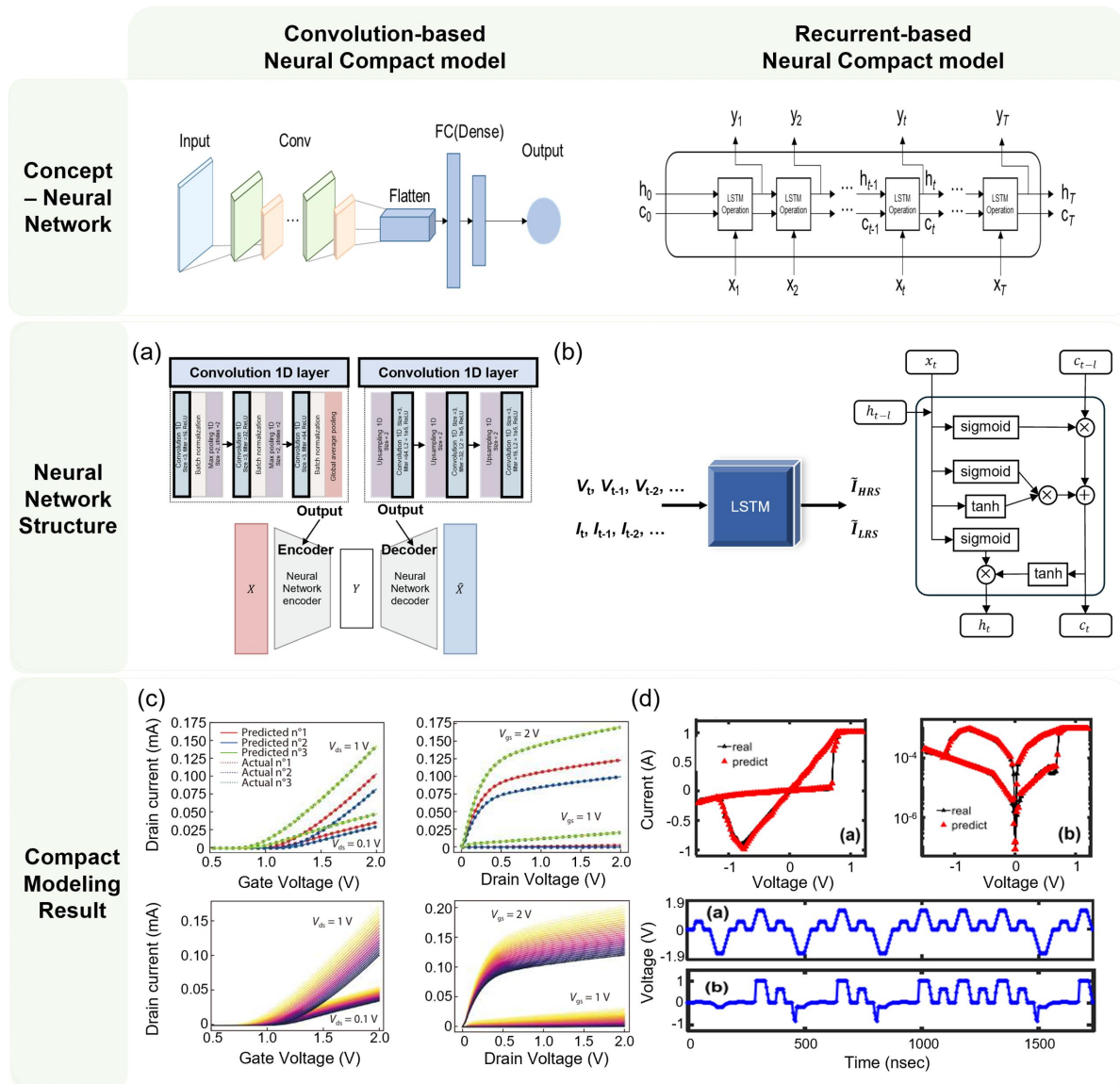


FIG. 5. Demonstration of deep learning-based neural compact models utilizing spatial and temporal learning: (left) a convolution-based model and (right) a recurrent model using LSTM. (a) shows a 1D convolutional encoder-decoder structure applied to predict I-V characteristics across wide operating conditions. The modeling performance³⁷ is shown in (c). (b) presents the structure of an LSTM-based model,⁷¹ which captures time-series dependencies such as hysteresis or memory effects; resulting dynamic current and voltage behaviors³⁴ are shown in (d). Together, these models demonstrate the strength of deep architectures in modeling both static and dynamic responses in compact modeling tasks. (c) Reproduced with permission from Hirtz *et al.*, J. Semicond. **42**, 124101 (2021). Copyright 2021 Chinese Institute of Electronics. (d) Reproduced with permission from Lin *et al.*, IEEE Trans. Electron Devices **69**, 1835–1841 (2022). Copyright 2022 IEEE.

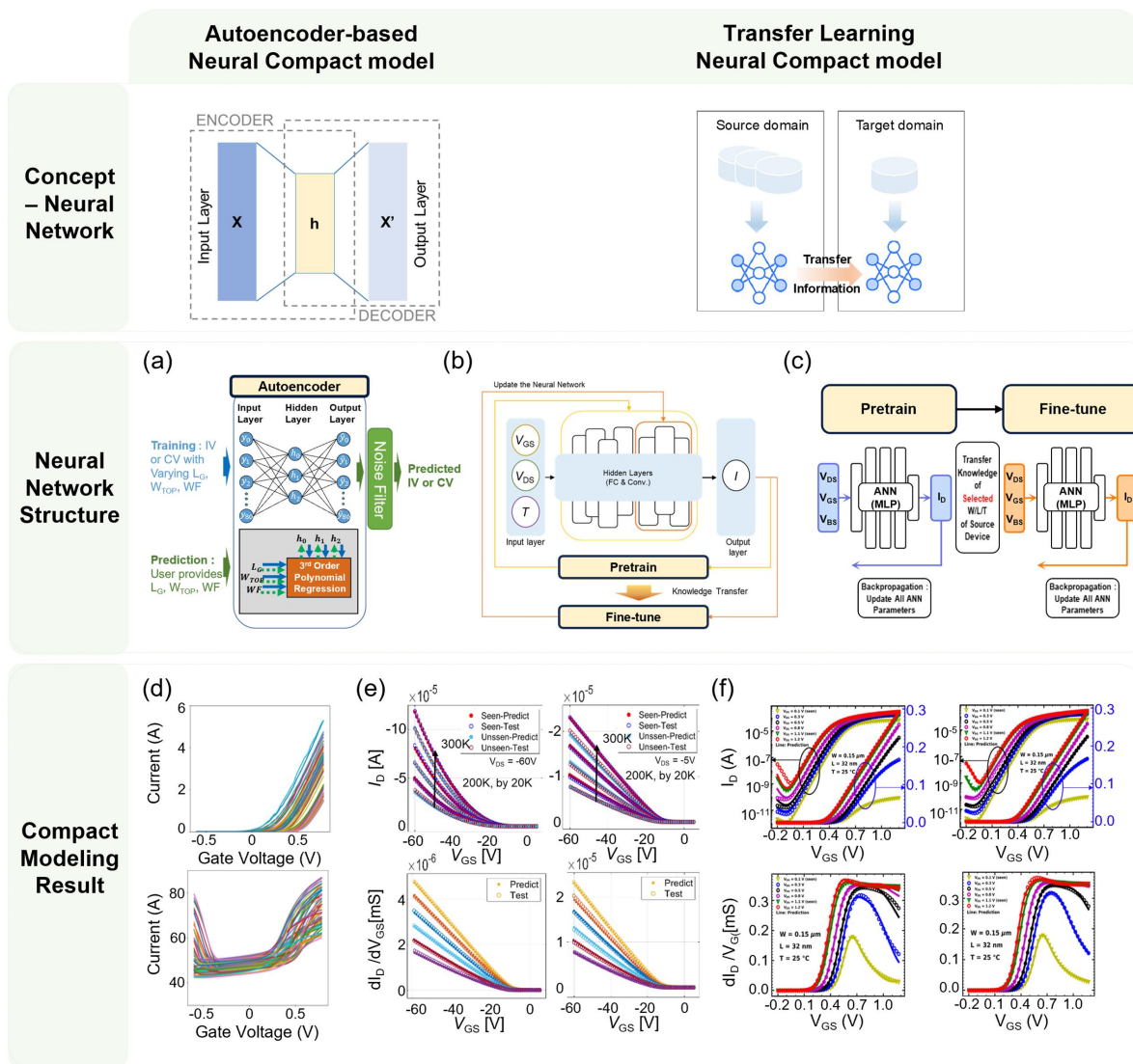


FIG. 6. Examples of deep neural compact models applied to capture more complex device behavior: (left) an autoencoder-based model, and (right) a transfer learning-based model. (a) shows the autoencoder architecture with polynomial regression applied to the latent space for predicting I–V or C–V curves; corresponding results⁷⁰ are shown in (d). (b) and (c) describe two implementations of transfer learning: both involve pretraining on source domain data and fine-tuning on target domain data, enabling knowledge transfer across device types. Compact modeling results for various temperatures⁶⁷ and bias conditions⁶⁸ are shown in (e) and (f), demonstrating the models' ability to generalize and accurately predict unseen data under changing environments. (d) Adapted with permission from Mehta and Wong, IEEE Electron Device Lett. **42**, 136–139 (2021). Copyright 2021 Authors, licensed under a Creative Commons Attribution (CC BY) license. (f) Reproduced with permission from Cha *et al.*, Solid-State Electron. **198**, 108450 (2022). Copyright 2022 Elsevier.

key training parameters, accuracy metrics, and simulation environments to provide a comprehensive view of implementation practices (Figs. 5 and 6).

1. CNN-based neural compact model

CNNs are NNs that mimic the structure of the human visual system, capable of learning filters to generate feature maps. They have demonstrated excellent performance in the field of computer

vision. Specifically, CNNs excel at pattern recognition and object detection tasks, making them well-suited for image recognition. Therefore, CNNs have also proven to be efficient in analyzing device data by extracting and classifying features from patterns and signals, thereby showcasing their effectiveness in device data analysis.

A CNN-based compact modeling framework was developed using TCAD-generated multi-I–V datasets to address data scarcity and accelerate device characterization.³⁷ Instead of relying on

reduced-order representations or predefined figures of merit, raw electrical characteristics were directly utilized as input, allowing convolution layers to capture local and hierarchical patterns across voltage sweeps. Both inverse and forward modeling tasks were performed: process/device parameters were inferred from electrical responses, and conversely, multi-output I - V characteristics were predicted from structural variables. A deconvolutional CNN was employed for forward modeling, while a standard 1D CNN architecture was used for reverse mapping. Prediction accuracy reached R^2 _score values between 0.982 and 0.999 for key parameters such as oxide thickness and doping concentration. Compared to traditional TCAD simulations, the trained model yielded an inference speedup of over 9000 $\times$. The proposed method demonstrated strong generalization on unseen samples, without requiring prior dimensionality reduction. The ability to model full multi-output behavior with minimal preprocessing highlights the suitability of CNNs for efficient, scalable, and data-rich compact modeling, particularly for advanced semiconductor nodes where simulation cost and data availability are critical limitations.

These results confirm the effectiveness of CNNs in compact modeling tasks that require both accuracy and scalability. By leveraging spatial feature extraction and enabling high-throughput inference, CNN-based frameworks offer a promising path toward replacing or complementing conventional physics-based simulations in next-generation semiconductor design flows.

2. RNN-based neural compact model

RNNs are a class of deep learning models designed to handle sequential data, where past inputs influence current predictions. Unlike traditional feedforward neural networks, which process data independently, RNNs incorporate feedback loops that allow them to retain information from previous time steps. This makes them particularly well-suited for applications such as time-series forecasting, speech recognition, and sequential decision-making. However, standard RNNs suffer from vanishing gradient problems, which limit their ability to capture long-term dependencies. To address this, advanced architectures like LSTM networks introduce gated mechanisms that selectively retain or forget information, making them highly effective for tasks requiring memory over extended sequences.

One research effectively captures the I - V characteristics of RRAM devices, learning both low resistance state (LRS) and high resistance state (HRS) within a single framework without requiring predefined transition rules. Unlike traditional compact models that use separate equations for each state, the LSTM model dynamically learns resistive switching behavior through its memory retention capability, enabling a unified representation. Trained on I - V measurements from RRAM devices, the model incorporates process variations from post-oxide annealing, allowing it to recognize and adjust to fabrication-induced fluctuations. By including both sinusoidal and random walk voltage sweeps, the dataset ensures robustness to non-ideal voltage waveforms, making the model suitable for real-world circuit simulations.

The LSTM network architecture follows a time-series learning approach, where prior voltage states influence current predictions. The model consists of a single LSTM layer with recurrent connections and a fully connected output layer, utilizing tanh and Sigmoid activation functions to capture nonlinear memory effects. Trained with the

Adam optimizer and RMSE loss, the model achieves smooth gradient updates and fast convergence, making it a highly efficient solution for compact memory device modeling. The LSTM-based compact model demonstrates superior predictive accuracy across multiple datasets, achieving an RMSE of 0.0096 for sinusoidal input voltage data and 0.0148 for random walk input data, highlighting its ability to generalize across diverse input sequences. When process variations (annealing conditions) are incorporated, the model achieves an RMSE of 0.0028, confirming its adaptability while maintaining near-SPICE accuracy.

A key advantage of this approach is its capability to accurately capture I - V shifts caused by fabrication variations, significantly reducing the need for manual tuning of compact model parameters. By incorporating annealing temperature as an additional input feature, the model effectively predicts device performance across different process conditions without requiring retraining, making it highly suitable for process optimization and yield enhancement in semiconductor manufacturing. Compared to conventional compact models that rely on empirical equations and extensive parameter extraction, the LSTM-based model offers a unified representation of switching behavior, eliminating the need for separate equations for LRS and HRS. Furthermore, its data efficiency, requiring only 400 training samples per I - V curve, makes it a practical and scalable alternative for data-limited semiconductor modeling scenarios.

This LSTM-based approach underscores the growing importance of sequence-aware architectures in compact modeling of memory devices. By replacing traditional equation-based modeling with a data-driven, unified framework, it enables accurate, process-adaptive prediction of resistive switching behavior with minimal data requirements. The model's ability to generalize across voltage waveforms and fabrication conditions not only enhances modeling fidelity but also supports integration into circuit-level simulations. Recent studies show that RNNs are increasingly adopted in neural compact models when device behavior exhibits strong hysteresis or transient dependence that cannot be captured by static mappings. In GRU (Gated Recurrent Unit)-based compact modeling of feedback field effect transistors,⁷² recurrent states enable accurate learning of latch type dynamics from voltage sweep sequences, and the resulting models are shown to be compatible with circuit level simulation while achieving substantial speed improvement over TCAD based approaches. Similarly, recent RRAM modeling work demonstrates that RNN style architectures can represent history dependent switching behavior and variability under transient excitation, while highlighting the importance of stable state evolution for reliable simulation use.⁷³ Together, these results indicate a growing role of sequence aware neural networks in compact modeling of emerging devices where hysteresis and transient effects are intrinsic to operation.

As the complexity and variability of emerging memory technologies continue to grow, such temporal learning frameworks offer a scalable, interpretable, and simulation-compatible path forward for compact modeling in next-generation semiconductor design.

3. Autoencoder-based neural network

Autoencoders (AEs) are neural network architectures that compress high-dimensional input data into a lower-dimensional latent space and then reconstruct the original data from this compact representation. Unlike conventional supervised models, AEs learn without

explicit labels, making them particularly suitable for compact modeling tasks where labeled datasets are limited. In semiconductor device modeling, this structure enables the extraction of essential physical behavior from complex electrical characteristics without requiring prior domain-specific feature engineering. Leveraging this property, AE-based approaches can efficiently learn underlying device trends and structure–property relationships directly from raw I – V and C – V curves, making them promising tools for modeling in data-constrained scenarios.

A novel methodology applies AE-based manifold learning to model FinFET behavior, achieving high accuracy even with as few as 25–50 data samples.⁷⁰ By leveraging low-dimensional latent space learning, the model effectively extracts meaningful representations of I – V and C – V curves without requiring explicit feature engineering. The AE architecture is designed to map latent variables (h_2 , h_2 , h_2) to key physical parameters such as gate length, fin width, and work function using third-order polynomial regression. This approach allows for precise I – V and C – V predictions without requiring direct physical intuition, making it a powerful tool for device modeling in data-scarce scenarios.

The model was trained on three different dataset variations (AE200, AE50, and AE25) to assess its performance under different data availability conditions. AE200, trained with 200 samples, accurately reproduces the entire I – V and C – V curves. AE50, trained with 50 samples, maintains the primary patterns despite minor accuracy degradation. Even with only 25 samples, AE25 successfully captures the key trends of I – V and C – V characteristics, demonstrating the robustness of the AE-based approach in low-data regimes. The AE consists of a five-layer NN with a structure of 40-40-3-40-40 neurons, where the latent space contains three neurons. It was trained using the Adam optimization algorithm with MSE loss, ensuring efficient convergence and high accuracy in reconstructing device characteristics. This study highlights the potential of AE-based latent space learning as a compact and efficient modeling technique for semiconductor devices, enabling accurate characterization with significantly reduced data requirements. Recent work shows that autoencoder-based neural networks are increasingly used to improve data efficiency and generalization in neural compact modeling. Large scale encoder–decoder frameworks leverage latent representations learned from extensive pretraining to enable accurate I – V and C – V prediction even under sparse measurement conditions, effectively supporting rapid adaptation to new technology nodes.⁷⁴ More recent studies further focus on enhancing interpretability by enforcing disentangled latent spaces, where individual latent dimensions are shown to correlate with physically meaningful device characteristics such as threshold voltage and short-channel effects.⁷⁵ Together, these results indicate a growing trend toward autoencoder-based compact models that balance predictive accuracy, data efficiency, and physical interpretability for reliable DTCO applications.

This AE-based compact modeling framework demonstrates the effectiveness of unsupervised latent space learning for the characterization, particularly under severe data constraints. By learning to represent complex device behavior in a compact three-dimensional manifold, the model accurately predicts electrical characteristics with as few as 25 training samples. The use of polynomial regression in the latent space further enables interpretable parameter mapping without requiring detailed physical modeling. Overall, this methodology offers

a scalable and data-efficient alternative to traditional compact modeling techniques, opening new opportunities for fast and accurate device modeling in early-stage design or limited-measurement environments.

4. Transfer learning compact model

To address the data scarcity commonly encountered in next-generation semiconductor device modeling, a knowledge transfer-based compact modeling framework was introduced.⁶⁸ This approach aims to enhance the performance of ANN-based compact models when only a limited amount of measurement data are available for the target device. Rather than relying solely on device-specific datasets, the model leverages rich data from similar or prior-generation devices, transferring learned representations via transfer learning and meta learning techniques.

In the transfer learning pipeline, an MLP neural network is first pretrained on a large dataset from a well-characterized source technology node. This pretrained model is then fine-tuned using a limited dataset from the target node. By adapting weights through backpropagation, the ANN successfully captures complex device phenomena such as gate-induced drain leakage (GIDL) that are otherwise missed by models trained from scratch. Compared to randomly initialized models, the fine-tuned model showed significantly reduced errors (linear error: 4.3% vs 22.9%) and faster convergence. Furthermore, training time is reduced from 538 to 186 s per $W/L/T$ condition.

Complementing this, the meta learning strategy employs an encoder–decoder architecture trained across multiple $W/L/T$ conditions from the source dataset. During meta-training, the model learns how to adapt to new $W/L/T$ conditions using minimal support data, enabling fast and accurate predictions without full retraining. This method achieved even lower test errors (linear: 2.3%, log: 0.11%) and the shortest adaptation time (1 s), outperforming both transfer learning and conventional training. Notably, the meta-trained model maintained excellent performance across a wide range of electrical parameters including $I_{D,LIN}$, $I_{D,SAT}$, V_{TH} , and GIDL current, even with reduced data availability.

This study demonstrates the efficacy of transferring latent knowledge—either through pretrained model weights or learned adaptation strategies limited regimes. By embedding physics implicitly through learned representations, the framework reduces reliance on extensive parameter extraction and provides a flexible modeling path for emerging technology nodes.

Another research result proposed a neural compact model for organic field-effect transistors (OFETs) incorporating transfer learning to improve accuracy and reduce training time.⁶⁷ Traditional compact models, whether analytical or lookup-table-based, required extensive calibration and often struggled with nonlinear behaviors arising from material properties and device structure. To address these challenges, the authors developed a data-driven model using deep learning, specifically a CNN, and demonstrated the advantages of transfer learning for efficient model adaptation. To construct the dataset, the study utilized TCAD simulations calibrated against experimental measurements. The dataset included I – V characteristics of OFETs exhibiting Gaussian disorder, capturing the influence of temperature, charge carrier concentration, and electric field dependence on mobility. Principal component analysis (PCA) was performed to

validate the similarity between TCAD and experimental data, ensuring that the ML model could generalize across different conditions.

The neural compact model consisted of a seven-layer architecture, including fully connected (FC) layers and convolutional layers to extract essential features from input variables V_{GS} , V_{DS} , and temperature (T). Unlike traditional physics-based models, this approach learned the complex transport physics without requiring predefined equations. The base model achieved the MSE of 0.0354 and a R^2 score value of 0.998, demonstrating its capability to accurately predict transistor behavior.

By freezing early layers responsible for general feature extraction and fine-tuning only the final fully connected layers, the model achieved high prediction accuracy with minimal data. The approach demonstrated mean absolute percentage errors of 4.32% and 7.28% for transfer across temperature and drain voltage domains, respectively, while preserving SPICE-relevant properties such as transconductance continuity and differentiability. The training time was reduced by approximately 50% compared to full retraining, and stable performance was maintained even with as few as 20 training samples. Circuit-level validation further confirmed the practicality of this method, with inverter simulations showing less than 5% deviation from TCAD references. These results highlight the effectiveness of transfer learning not only in scaling compact modeling across process conditions but also in modeling disordered or emerging device structures where physical equations are difficult to derive.

IV. INSIGHTS AND TRAJECTORIES IN NEURAL COMPACT MODEL FOR NEXT-GENERATION SEMICONDUCTOR DESIGN

Previous review studies on AI-based device modeling have primarily focused on presenting the overall landscape of machine-learning-assisted compact modeling, including algorithmic frameworks, parameter extraction techniques, and representative applications across different device technologies.^{76–79} While these studies provide valuable conceptual overviews, discussions on the practical

implementation of neural compact models such as neural network architecture configuration, hyperparameter selection, dataset construction, and computational requirements remain relatively limited. In contrast, this review aims to provide a more implementation-oriented perspective by systematically analyzing quantitative aspects reported in prior studies, including network depth, optimizer choice, training strategies, and computational environments summarized in tables.

Neural compact models have shown great potential as an alternative to traditional compact modeling methods for transistors and logic design. As discussed in Secs. III A – III C, these models must simultaneously address two key perspectives. From the compact model perspectives, they should remain simple, computationally efficient, physically meaningful, and accurate. From the machine learning standpoint, they must be supported by well-structured data pipelines, including data acquisition, preprocessing, network architecture selection, hyperparameter tuning, optimization methods, and model evaluation. The studies reviewed in this paper reflect this balance, showing that neural networks can successfully perform function approximation without relying on complex analytical equations, thereby simplifying modeling tasks and significantly reducing simulation time. These models also demonstrate high prediction accuracy while incorporating physically relevant factors, such as device-specific architecture choices, physics-informed constraints, and knowledge transfer, especially for emerging semiconductor technologies.

However, several limitations have been identified that could impact the reproducibility and scalability of these models. Many existing studies do not provide sufficient details on critical training components. In particular, many studies do not specify how much training data were used or whether the data originated from simulations or measurements. Descriptions of preprocessing methods, neural network configurations, and hyperparameter settings are also inconsistently reported. These omissions limit the ability of other researchers to replicate or build upon existing work. Future studies should aim to provide comprehensive documentation of the learning

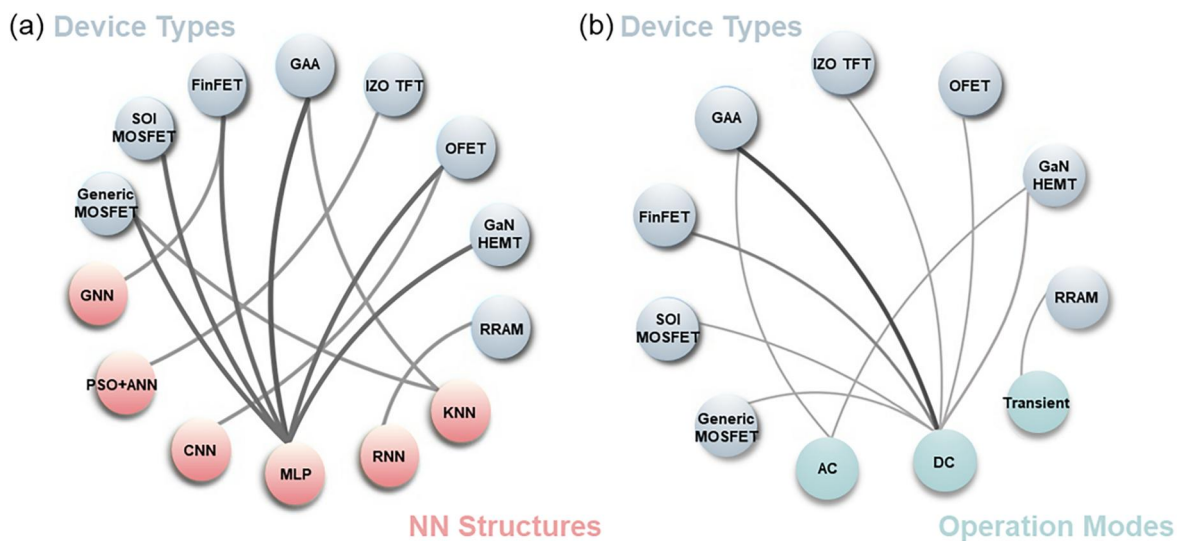


FIG. 7. Frequency mapping of neural compact models for devices: device types, neural network (NN) structures, and operation modes.

setup to promote transparency and more consistent benchmarking across the field.

In this context, Fig. 7 presents a visual mapping of neural compact model research to date. Figure 7(a) shows the associations between different types of transistors and the NN architectures applied in their modeling, while Fig. 7(b) illustrates the relationships between transistor types and the specific electrical characteristics modeled. Thicker lines in both diagrams denote higher application frequency or stronger relevance. This mapping highlights a key trend: the selection of a neural network model is often tailored to the nature of the target device. For example, passive devices or relatively simple transistor structures are frequently modeled using MLPs due to their efficiency and sufficient expressive capacity under low-complexity conditions. On the other hand, active devices with highly nonlinear or time-dependent behaviors, such as RRAM or ferroelectric devices, often require deeper or more specialized architectures such as RNNs, CNNs, or LSTM variants. Similarly, modeling more complex outputs such as both I - V and C - V characteristics increases the dimensionality of the problem, requiring networks with greater depth and generalization capability.

From a modeling perspective, each neural network architecture presents distinct advantages and limitations depending on the characteristics of the target device and the operating regime. MLP-based models remain the most widely adopted due to their structural simplicity, fast training, and flexibility in representing general device behaviors across various transistor technologies. However, their capability to capture device-specific dynamics such as transient switching behavior, hysteresis, or strong temporal dependence can be limited. To address these phenomena, more specialized architectures such as RNNs or CNNs have been explored, as they are better suited to capturing temporal correlations or structured nonlinear patterns in device responses. Nevertheless, incorporating such architectures typically increases model complexity and computational cost. As a result, careful consideration of model architecture, training strategy, and dataset structure becomes increasingly important, particularly when dealing with high-dimensional device characteristics or dynamic operating regimes.

Moreover, this trend reflects a deeper insight into how the physical characteristics of transistors influence network selection. Devices requiring transient modeling, such as those with switching memory behavior, benefit from RNNs that can retain temporal sequence information. In contrast, devices that exhibit spatially variant behaviors—due to layout-dependent parasitics or complex geometry—are more effectively modeled by CNNs, which can capture local and hierarchical patterns. Consequently, neural network selection is not arbitrary but driven by the underlying physical and electrical traits of the device.

In addition, ensuring compatibility with SPICE simulation environments remains critical for practical deployment. Neural models must be convertible into formats such as Verilog-A, delivering robust and accurate predictions across a range of conditions. While several studies have demonstrated this compatibility, future work should expand on this by providing benchmarks on computational efficiency, hardware requirements, and simulation accuracy. For instance, specifying CPU/GPU (Graphics Processing Unit) configurations, training durations, and inference latency would offer valuable context for deployment feasibility, particularly in industry where resource constraints are common.

Taken together, these findings suggest that neural compact model development must not only emphasize predictive performance but also align network architecture with device-specific modeling objectives. The visual synthesis in Fig. 7 offers practical guidance for selecting appropriate NN structures based on device behavior and modeling scope, pointing toward more informed and physically grounded modeling strategies. This approach can help define clearer design guidelines and support the next generation of compact modeling frameworks.

Looking ahead, future research directions should include extending neural compact model development to a wider range of emerging semiconductor devices. Current efforts have largely focused on a limited set of transistors such as FinFETs, GAAFETs, and RRAM. However, other devices like ferroelectric FETs, tunnel FETs, and negative capacitance FETs, as well as transistors based on 2D materials and organic semiconductors, are gaining attention due to their novel properties. These devices exhibit unique electrical behaviors such as hysteresis, abrupt switching, and ambipolar conduction, which are often difficult to model using traditional physics-based formulations. Neural networks offer the flexibility to learn such complex characteristics directly from data, making them an ideal tool for compact modeling in cases where analytical models are insufficient or unavailable.

Another key consideration is the integration of neural compact models into mainstream design environments. While several implementations have demonstrated SPICE compatibility through Verilog-A descriptions, the broader adoption of AI-based models in conventional EDA tools is still in its early stages. Some workflows may evolve toward co-simulation platforms or data-driven simulation frameworks that natively support machine learning models and offer greater flexibility for retraining or online adaptation. This shift could potentially give rise to a new generation of simulation tools designed specifically around AI-enhanced compact models, particularly as the design of emerging devices increasingly relies on data-centric methods.

Establishing consistent deployment strategies and benchmarking standards will be essential for enabling the widespread use of neural compact models. With clearer guidelines for data acquisition, training procedures, model evaluation, and hardware requirements, the field can progress toward more reproducible, efficient, and scalable modeling practices. This foundation will be critical not only for advancing academic research but also for supporting industrial adoption in next-generation semiconductor design and manufacturing.

V. CONCLUSION

Neural compact models have emerged as a powerful and versatile alternative to conventional approaches for semiconductor circuit simulation, especially in the context of advanced and emerging device technologies. By employing a range of neural architectures—including shallow MLPs, deep CNNs, RNNs, and advanced learning frameworks—these models can accurately approximate complex device behaviors while substantially reducing development time and computational cost. The studies reviewed here illustrate the capability of neural networks to capture nonlinear DC, AC, and transient characteristics across a broad spectrum of passive and active devices, such as antennas, generic MOSFETs, SOI MOSFETs, FinFETs, GAAFETs, TFTs, GaN HEMTs, and RRAMs. Furthermore, the incorporation of domain knowledge through physics-informed learning, prior knowledge embedding, and hybrid modeling approaches has

significantly enhanced both the physical rigorousness and generalization performance of neural compact models.

Despite notable progress, several critical challenges remain. Key issues such as reproducibility, transparency in training data and methodologies, and the lack of standardized evaluation protocols hinder the broader adoption of neural compact models. Future research should prioritize the development of consistent benchmarking practices, comprehensive reporting of hardware configurations and dataset characteristics, and the integration of explainable AI techniques to enhance model interpretability to provide feedback for further device technology development. Moreover, achieving robust cross-device generalization and enabling real-time inference (i.e., circuit simulation) will be vital for practical deployment in circuit- and system-level applications. As advancements continue in both machine learning and semiconductor device modeling, neural compact models are expected to play a pivotal role in device modeling for circuit simulation, which could potentially accelerating adoption of emerging device technologies for next-generation electronic systems.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Minsun Cho and Deokho Jang contributed equally to this paper.

Minsun Cho: Formal analysis (equal); Funding acquisition (supporting); Investigation (equal); Methodology (equal); Software (equal); Visualization (equal); Writing – original draft (equal). **Deokho Jang:** Formal analysis (equal); Investigation (equal); Methodology (equal); Software (equal); Visualization (equal); Writing – original draft (equal). **O-Joun Lee:** Conceptualization (supporting); Investigation (supporting); Supervision (supporting); Writing – review & editing (supporting). **Chang-Hyun Kim:** Conceptualization (supporting); Investigation (supporting); Supervision (supporting); Writing – review & editing (supporting). **Sungyeop Jung:** Conceptualization (lead); Funding acquisition (lead); Project administration (lead); Resources (lead); Supervision (lead); Validation (lead); Writing – review & editing (lead).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

REFERENCES

- ¹V. T. Hoang, H.-J. Jeon, E.-S. You, Y. Yoon, S. Jung, and O.-J. Lee, “Graph representation learning and its applications: A survey,” *Sensors* **23**, 4168 (2023).
- ²P. Raghuvanshi, “Revolutionizing semiconductor design and manufacturing with AI,” *J. Knowl. Learn. Sci. Technol.* **3**, 272–277 (2024).
- ³P. Sengupta, A. Tyagi, Y. Chen, and J. Hu, “How good is your verilog RTL code? A quick answer from machine learning,” in *Proceedings of the 41st IEEE/ACM International Conference on Computer-Aided Design* (IEEE, 2022), pp. 1–9.
- ⁴Y.-C. Lu, S. Nath, V. Khandelwal, and S. K. Lim, “Doomed run prediction in physical design by exploiting sequential flow and graph learning,” in *IEEE/ACM International Conference on Computer Aided Design (ICCAD)* (IEEE, 2021), pp. 1–9.
- ⁵Y. Li, Y. Lin, M. Madhusudan, A. Sharma, W. Xu, S. S. Sapatnekar, R. Harjani, and J. Hu, “A customized graph neural network model for guiding analog IC placement,” in *Proceedings of the 39th International Conference on Computer-Aided Design* (IEEE, 2020), pp. 1–9.
- ⁶A. Mirhoseini, A. Goldie, M. Yazgan, J. W. Jiang, E. Songhori, S. Wang, Y.-J. Lee, E. Johnson, O. Pathak, A. Nova et al., “A graph placement methodology for fast chip design,” *Nature* **594**, 207–212 (2021).
- ⁷H. Wang, K. Wang, J. Yang, L. Shen, N. Sun, H.-S. Lee, and S. Han, “GCN-RL circuit designer: Transferable transistor sizing with graph neural networks and reinforcement learning,” in *57th ACM/IEEE Design Automation Conference (DAC)* (IEEE, 2020), pp. 1–6.
- ⁸J. Hu and A. B. Kahng, “Invited paper: The inevitability of AI infusion into design closure and signoff,” in *IEEE/ACM International Conference on Computer Aided Design (ICCAD)* (Institute of Electrical and Electronics Engineers, Inc., 2023), pp. 1–7.
- ⁹Z. Chen, J. Li, and Y. Xu, “Machine learning-assisted high-throughput semi-empirical search of OFET molecular materials,” *arXiv:2107.02613* (2021).
- ¹⁰Y. Ji and L. Chen, “A fine-grained, end-to-end feature-scale CMP modeling paradigm based on fully convolutional neural networks,” *IEEE Trans. Semicond. Manuf.* **36**, 231–238 (2023).
- ¹¹Q. Zhang, H. Huang, J. Li, Y. Zhang, and Y. Li, “CmpCNN: CMP modeling with transfer learning CNN architecture,” *ACM Trans. Des. Autom. Electron. Syst.* **28**, 1–18 (2023).
- ¹²S. Shim, S. Choi, and Y. Shin, “Machine learning (ML)-based lithography optimizations,” in *IEEE Asia Pacific Conference on Circuits and Systems (APCCAS)* (IEEE, 2016), pp. 530–533.
- ¹³Y. Watanabe, T. Kimura, T. Matsunawa, and S. Nojima, “Accurate lithography simulation model based on convolutional neural networks,” *Proc. SPIE* **10147**, 137–145 (2017).
- ¹⁴M. Phute, A. Sahastrabudhe, S. Pimparkhede, S. Potphode, K. Rengade, and S. Shilaskar, “A survey on machine learning in lithography,” in *International Conference on Artificial Intelligence and Machine Vision (AIMV)* (IEEE, 2021), pp. 1–6.
- ¹⁵I. Kim, S. J. Park, C. Jeong, M. Shim, D. S. Kim, G.-T. Kim, and J. Seok, “Simulator acceleration and inverse design of fin field-effect transistors using machine learning,” *Sci. Rep.* **12**, 1140 (2022).
- ¹⁶D. Jang, S. Kim, and J. Kim, “Deep-learning-based inverse design of colloidal quantum dots,” *Opt. Commun.* **559**, 130384 (2024).
- ¹⁷C. Gallo and V. Capozzi, “A wafer bin map ‘relaxed’ clustering algorithm for improving semiconductor production yield,” *Open Comput. Sci.* **10**, 231–245 (2020).
- ¹⁸Y.-C. Ko and H. Fujita, “An evidential analytics for buried information in big data samples: Case study of semiconductor manufacturing,” *Inf. Sci.* **486**, 190–203 (2019).
- ¹⁹B. Schulz, C. Jacobi, A. Gisbrecht, A. Evangelos, C. W. Chan, and B. P. Gan, “Graph representation and embedding for semiconductor manufacturing fab states,” in *Winter Simulation Conference (WSC)* (IEEE, 2022), pp. 3382–3393.
- ²⁰G. Gildenblat, *Compact Modeling: Principles, Techniques and Applications* (Springer, The Netherlands, 2010).
- ²¹K. Hornik, M. Stinchcombe, and H. White, “Multilayer feedforward networks are universal approximators,” *Neural Networks* **2**, 359–366 (1989).
- ²²V. Litovski, J. Radjenović, Ž. Mrčarić, and S. L. Milenković, “Mos transistor modelling using neural network,” *Electron. Lett.* **28**, 1766–1768 (1992).

- ²³J. X. Xu, M. C. Yagoub, R. Ding, and Q. J. Zhang, "Exact adjoint sensitivity analysis for neural-based microwave modeling and design," *IEEE Trans. Microwave Theory Tech.* **51**, 226–237 (2003).
- ²⁴K. Nishiguchi, T. Kawasaki, and M. Tanomura, "Neural network based GaN HEMT modelling for millimeter wave power amplifiers," in *IEEE/MTT-S International Microwave Symposium (IMS 2022)* (IEEE, 2022), pp. 382–385.
- ²⁵J. Xu, D. Gunyan, M. Iwamoto, A. Cognata, and D. E. Root, "Measurement-based non-quasi-static large-signal FET model using artificial neural networks," in *IEEE MTT-S International Microwave Symposium Digest* (IEEE, 2006), pp. 469–472.
- ²⁶J. Xu, S. Halder, F. Kharabi, J. McMacken, J. Gering, and D. E. Root, "Global dynamic FET model for GaN transistors: DynaFET model validation and comparison to locally tuned models," in *83rd ARFTG Microwave Measurement Conference* (IEEE, 2014), pp. 1–6.
- ²⁷J. Xu, R. Jones, S. A. Harris, T. Nielsen, and D. E. Root, "Dynamic FET model—DynaFET—for GaN transistors from NVNA active source injection measurements," in *IEEE MTT-S International Microwave Symposium* (Institute of Electrical and Electronics Engineers, Inc., 2014), pp. 1–3.
- ²⁸A. H. Zaabab, Q.-J. Zhang, and M. Nakhla, "A neural network modeling approach to circuit optimization and statistical design," *IEEE Trans. Microwave Theory Tech.* **43**, 1349–1358 (1995).
- ²⁹A. Veluswami, M. S. Nakhla, and Q.-J. Zhang, "The application of neural networks to EM-based simulation and optimization of interconnects in high-speed VLSI circuits," *IEEE Trans. Microwave Theory Tech.* **45**, 712–723 (1997).
- ³⁰P. M. Watson and K. C. Gupta, "Design and optimization of CPW circuits using EM-ANN models for CPW components," *IEEE Trans. Microwave Theory Tech.* **45**, 2515–2523 (1997).
- ³¹G. L. Creech, B. J. Paul, C. D. Lesniak, T. J. Jenkins, and M. C. Calcaterra, "Artificial neural networks for fast and accurate EM-CAD of microwave circuits," *IEEE Trans. Microwave Theory Tech.* **45**, 794–802 (1997).
- ³²F. Wang and Q.-J. Zhang, "Knowledge-based neural models for microwave design," *IEEE Trans. Microwave Theory Tech.* **45**, 2333–2343 (1997).
- ³³P. Watson, K. Gupta, and R. Mahajan, "Applications of knowledge-based artificial neural network modeling to microwave components," *Int. J. RF Microwave Comput.-Aided Eng.* **9**, 254–260 (1999).
- ³⁴J. W. Bandler, M. A. Ismail, J. E. Rayas-Sánchez, and Q.-J. Zhang, "Neuromodeling of microwave circuits exploiting space-mapping technology," *IEEE Trans. Microwave Theory Tech.* **47**, 2417–2427 (1999).
- ³⁵Y. Harkouss, J. Rousset, H. Chehade, E. Ngoya, D. Barataud, and J.-P. Teyssier, "The use of artificial neural networks in nonlinear microwave devices and circuits modeling: An application to telecommunication system design (invited article)," *Int. J. RF Microwave Comput.-Aided Eng.* **9**, 198–215 (1999).
- ³⁶J. Xu, J. Horn, M. Iwamoto, and D. E. Root, "Large-signal FET model with multiple time scale dynamics from nonlinear vector network analyzer data," in *IEEE MTT-S International Microwave Symposium* (IEEE, 2010).
- ³⁷T. Hirtz, S. Huurman, H. Tian, Y. Yang, and T.-L. Ren, "Framework for TCAD augmented machine learning on multi-I-V characteristics using convolutional neural network and multiprocessing," *J. Semicond.* **42**, 124101 (2021).
- ³⁸S. Woo, H. Jeong, J. Choi, H. Cho, J.-T. Kong, and S. Kim, "Machine-learning-based compact modeling for sub-3-nm-node emerging transistors," *Electronics* **11**, 2761 (2022).
- ³⁹Z. Yang, A. D. Gaidhane, K. Anderson, G. Workman, and Y. Cao, "Graph-based compact model (GCM) for efficient transistor parameter extraction: A machine learning approach on 12 nm FinFETs," *IEEE Trans. Electron Devices* **71**, 254–262 (2024).
- ⁴⁰S. Hatami, M. Y. Azizi, H.-R. Bahrami, D. Motavallizadeh, and A. Afzali-Kusha, "Accurate and efficient modeling of SOI MOSFET with technology independent neural networks," *IEEE Trans. Comput.-Aided Des. Integr. Circuits Syst.* **23**, 1580–1587 (2004).
- ⁴¹F. Klemme, J. Prinz, V. M. van Santen, J. Henkel, and H. Amrouch, "Modeling emerging technologies using machine learning: Challenges and opportunities," in *Proceedings of the 39th International Conference on Computer-Aided Design* (IEEE, 2020), pp. 1–9.
- ⁴²C.-T. Tung, M.-Y. Kao, and C. Hu, "Neural network-based and modeling with high accuracy and potential model speed," *IEEE Trans. Electron Devices* **69**, 6476–6479 (2022).
- ⁴³R. Li, H. Luo, Y. Wang, Z. Yuan, A. Asenov, X. Miao, and X. Wang, "Improved compact model extraction of statistical variability in 5 nm nano-sheet transistors and applied to SRAM simulations," *Semicond. Sci. Technol.* **37**, 095010 (2022).
- ⁴⁴L. E. Calvet, S. El-Nakouzi, Z. Li, Y. Kim, A. Zaibi, P. Golec, I. M. Bhattacharyya, Y. Bonnassieux, L. Kadura, and B. Iniguez, "Simulating organic thin film transistors using multilayer perceptron regression models to enable circuit design," *Adv. Electron. Mater.* **10**, 2400515 (2024).
- ⁴⁵Y.-W. Ho, T. S. Rawat, Z.-K. Yang, S. Pratik, G.-W. Lai, Y.-L. Tu, and A. Lin, "Neuroevolution-based efficient field effect transistor compact device models," *IEEE Access* **9**, 159048 (2021).
- ⁴⁶P. Yang, W. Ming, J. Liang, I. Lütke, S. Berry, and K. Floros, "Hybrid data-driven modeling methodology for fast and accurate transient simulation of SiC MOSFETs," *IEEE Trans. Power Electron.* **37**, 440–451 (2022).
- ⁴⁷W. Deng, W. Zhang, Y. Peng, W. Wu, J. Huang, and Z. Luo, "Compact modeling of metal-oxide TFTs based on artificial neural network and improved particle swarm optimization," *J. Comput. Electron.* **20**, 1043–1049 (2021).
- ⁴⁸Q. Chen and G. Chen, "Artificial neural network compact model for TFTs," in *7th International Conference on Computer Aided Design for Thin-Film Transistor Technologies (CAD-TFT)* (IEEE, 2016).
- ⁴⁹Q. Yang, G. Qi, W. Gan, Z. Wu, H. Yin, T. Chen, G. Hu, J. Wan, S. Yu, and Y. Lu, "Transistor compact model based on multigradient neural network and its application in SPICE circuit simulations for gate-all-around Si cold source FETs," *IEEE Trans. Electron Devices* **68**, 4181–4188 (2021).
- ⁵⁰J. Wang, Y.-H. Kim, J. Ryu, C. Jeong, W. Choi, and D. Kim, "Artificial neural network-based compact modeling methodology for advanced transistors," *IEEE Trans. Electron Devices* **68**, 1318–1325 (2021).
- ⁵¹H. Habal, D. Tsonev, and M. Schweikardt, "Compact models for initial MOSFET sizing based on higher-order artificial neural networks," in *Proceedings of the ACM/IEEE Workshop on Machine Learning for CAD* (IEEE, 2020), pp. 111–116.
- ⁵²Y. Kim, S. Myung, J. Ryu, C. Jeong, and D. S. Kim, "Physics-augmented neural compact model for emerging device technologies," in *International Conference on Simulation of Semiconductor Processes and Devices (SISPAD)* (IEEE, 2020), pp. 257–260.
- ⁵³M.-Y. Kao, H. Kam, and C. Hu, "Deep-learning-assisted physics-driven MOSFET current-voltage modeling," *IEEE Electron Device Lett.* **43**, 974–977 (2022).
- ⁵⁴A. S. Lin, P.-N. Liu, S. Pratik, Z.-K. Yang, T. Rawat, and T.-Y. Tseng, "RRAM compact modeling using physics and machine learning hybridization," *IEEE Trans. Electron Devices* **69**, 1835–1841 (2022).
- ⁵⁵G. Qi, X. Chen, G. Hu, P. Zhou, W. Bao, and Y. Lu, "Knowledge-based neural network SPICE modeling for MOSFETs and its application on 2D material field-effect transistors," *Sci. China: Inf. Sci.* **66**, 122405 (2023).
- ⁵⁶J. Zhang, Y. Lv, S. Chang, H. Wang, J. He, and Q. Huang, "Prior knowledge input neural network method for GFET description," *J. Comput. Electron.* **15**, 911–918 (2016).
- ⁵⁷A. Chvála, L. Nagy, J. Marek, J. Priesol, D. Donoval, and A. Šatka, "Neural network for circuit models of monolithic InAlN/GaN NAND and NOR logic gates," in *14th International Conference on Design & Technology of Integrated Systems in Nanoscale Era (DTIS)* (IEEE, 2019), pp. 1–4.
- ⁵⁸L. Zhang and M. Chan, "Artificial neural network design for compact modeling of generic transistors," *J. Comput. Electron.* **16**, 825–832 (2017).
- ⁵⁹S. Xiao, B.-Z. Wang, X. Zhong, and G. Wang, "Wideband mobile antenna design based on artificial neural network models," *Int. J. RF Microwave Comput.-Aided Eng.* **13**, 316–320 (2003).
- ⁶⁰J.-H. Chen, F. Chavez, C.-T. Tung, S. Khandelwal, and C. Hu, "A single neural network global I-V and C-V parameter extractor for BSIM-CMG compact model," *Solid-State Electron.* **216**, 108898 (2024).
- ⁶¹J.-H. Chen, F. Chavez, C.-T. Tung, S. Khandelwal, and C. Hu, "A multi-stage neural network IV and C-V BSIM-CMG model global parameter extractor for advanced GAAFET technologies," *Solid-State Electron.* **226**, 109081 (2025).
- ⁶²S. Eom, H. Yun, H. Jang, K. Cho, S. Lee, J. Jeong, and R.-H. Baek, "Neural compact modeling framework for flexible model parameter selection with high accuracy and fast SPICE simulation," *Adv. Intell. Syst.* **6**, 2300435 (2024).
- ⁶³B. Peng, F. Zhang, W. Dai, H. Wu, G. Cheng, R. Wang, M. Chan, and L. Zhang, "PHIMO-NN: Compact modeling by fusing device physics and neural

- networks for one-shot parameterization," *IEEE Trans. Comput.-Aided Des. Integr. Circuits Syst.* **45**, 795 (2026).
- ⁶⁴Y. Lee, K. Kim, and J. Lee, "A compact memristor model based on physics-informed neural networks," *Micromachines* **15**, 253 (2024).
- ⁶⁵H. Jeong, J. Choi, Y. Kim, J.-T. Kong, and S. Kim, "Efficient neural network-based compact modeling for novel device structures using a multi-fidelity model and active learning," *Electronics* **13**, 4840 (2024).
- ⁶⁶K. Sheelvardhan, S. Guglani, M. Ehteshamuddin, S. Roy, and A. Dasgupta, "Machine learning augmented compact modeling for simultaneous improvement in computational speed and accuracy," *IEEE Trans. Electron Devices* **71**, 239–245 (2024).
- ⁶⁷M. Cho, M. Franot, O.-J. Lee, and S. Jung, "A neural compact model based on transfer learning for organic FETs with Gaussian disorder," *J. Mater. Chem. C* **12**, 16691–16700 (2024).
- ⁶⁸Y. S. Cha, J. Park, C. Park, S. Chong, C.-H. Kim, C.-S. Lee, I. Jeong, and H. Cho, "A novel methodology for neural compact modeling based on knowledge transfer," *Solid-State Electron.* **198**, 108450 (2022).
- ⁶⁹A. S. Lin, S. Pratik, J. Ota, T. S. Rawat, T.-H. Huang, C.-L. Hsu, W.-M. Su, and T.-Y. Tseng, "A process-aware memory compact-device model using long-short term memory," *IEEE Access* **9**, 3126–3139 (2020).
- ⁷⁰K. Mehta and H.-Y. Wong, "Prediction of FinFET current-voltage and capacitance-voltage curves using machine learning with autoencoder," *IEEE Electron Device Lett.* **42**, 136–139 (2021).
- ⁷¹A. S. Lin, S. Pratik, J. Ota, T. S. Rawat, T.-H. Huang, C.-L. Hsu, W.-M. Su, and T.-Y. Tseng, "A process-aware memory compact-device model using long-short term memory," *IEEE Access* **9**, 3126–3139 (2021).
- ⁷²S. Son, T. Lee, J. Jeon, G. Choe, and S. Woo, "ML-assisted compact modeling of feedback field-effect transistors using a gated recurrent unit for circuit simulation," *IEEE Access* **13**, 211497 (2025).
- ⁷³G. Hong, I. Huh, J. H. You, J. M. Choe, Y. Kim, and C. Jeong, "ML-driven compact models for RRAMs: Addressing variability and simulation efficiency," *IEEE Electron Device Lett.* **46**, 876 (2025).
- ⁷⁴C. Park, S. Lee, J. Park, K. Rim, J. Park, S. Cho, J. Jeon, and H. Cho, "Large-scale training in neural compact models for accurate and adaptable MOSFET simulation," *IEEE J. Electron Devices Soc.* **12**, 745–751 (2024).
- ⁷⁵C. Park, H. Cho, and J. Lee, "Enhancing interpretability of neural compact models: Toward reliable device modeling," *IEEE J. Electron Devices Soc.* **12**, 495–501 (2024).
- ⁷⁶P. Raut, D. K. Panda, and A. K. Goyal, "A comprehensive review on next-generation modelling and optimization for semiconductor devices," *IEEE Access* **13**, 123724 (2025).
- ⁷⁷X. Li, Z. Wu, G. Rzepa, M. Karner, H. Xu, Z. Wu, W. Wang, G. Yang, Q. Luo, L. Wang *et al.*, "Overview of emerging semiconductor device model methodologies: From device physics to machine learning engines," *Fundam. Res.* **5**, 2149 (2025).
- ⁷⁸L. Zhang, B. Peng, Y. Li, H. Liu, W. Dai, and R. Wang, "When device modeling meets machine learning: Opportunities and challenges," in *Proceedings of the ACM/IEEE International Symposium on Machine Learning for CAD* (IEEE, 2024), pp. 1–6.
- ⁷⁹C. C. McAndrew, A. J. Scholten, K. K. Gullapalli, Y. Chauhan, and K. Xia, "Benchmarks for SPICE modeling and parameter extraction based on AI/ML," *IEEE Trans. Electron Devices* **72**, 1551 (2025).